

LC Connect: Assistant Comparison Kit

Status: DRAFT, 2026-09-29 (assistant runs done 15:18–15:55 CEST; post-fix re-verification 22:17–22:57 CEST; ChatGPT + LC reruns 22:43–22:46 CEST). **Evening update:** the two *ChatGPT + LC* columns now use valid reruns made at 22:43–22:46 CEST (files [-v2](#)), after the account owner replaced his ChatGPT custom instructions, which had mentioned Vendo. The 15:51–15:55 CEST ChatGPT + LC runs are superseded and kept in Appendix A. The *ChatGPT* (without) columns remain the 15:28–15:31 CEST runs; see "ChatGPT baseline" in Section 1. A post-fix Claude + LC Czech Republic rerun stalled (Section 4.4); after a one-call-rule fix and a city backfill, a repeat rerun the same night succeeded (Section 4.5). All eight assistant runs behind the comparison in Sections 3–5 are complete. The first four ("Claude"/"ChatGPT" without the connector, plus the first *Claude + LC* attempt) were done 15:18–15:31 CEST. The first *Claude + LC* attempt found the connector switched on but **not called**, because the account had a personal standing instruction to use no connector other than Vendo; that instruction was then lifted and both *Claude + LC* runs were repeated (15:44–15:49 CEST, files suffixed [-v2](#)) with the connector actually invoked. LC Connect was also installed in the test ChatGPT account (there named "LC Connector") and both *ChatGPT + LC* runs were completed 15:51–15:55 CEST. All answers below are copied from the chats; nothing was invented or estimated. **Separately, after two server fixes landed the same day (commits [d16086e](#), [a0dfd53](#)), the Claude + LC Poland question was re-run at 22:17:59 CEST to verify the fix; that run, plus an intermediate broken check at 16:03 CEST, is added as Section 3.4.**

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1. Method

Purpose. To show, without embellishment, what a general-purpose AI assistant answers to a typical lead-research question with and without access to Laser Components' own research base through the LC Connect connector. If an assistant names companies that cannot be found or verified, we record that as the finding.

Questions (identical wording in every run):

- **Q1.** "List companies in Poland that use pulsed laser diodes (PLD, 905 nm or 1550 nm) in their products, for example laser rangefinders, LiDAR or proximity fuzes. For each company give: name, city, the product or application, and how confident you are."
- **Q2.** The same prompt, with "Poland" replaced by "the Czech Republic".

Why the Czech Republic for Q2. We counted leads per country in the LC Connect database on 2026-09-29. The best-covered European countries are Poland (14 leads with a country assigned), the Czech Republic (13) and the United Kingdom (8). **Germany has no leads in the research base.** It has never been researched, and it is the natural next step. We chose the Czech Republic because it is the best-covered European country after Poland.

The five columns:

COLUMN	WHAT IT MEANS
Claude	Claude.ai, new chat, LC Connect turned off in the chat's connector menu
Claude + LC	Claude.ai, new chat, LC Connect turned on ; tool permissions approved if asked
ChatGPT	ChatGPT, new chat, LC Connect app not selected for the chat
ChatGPT + LC	ChatGPT, new chat, LC Connect app selected ; tool permissions approved if asked (22:43–22:46 CEST reruns, -v2)

COLUMN	WHAT IT MEANS
LC database	Direct read-only query of the live LC Connect server (<code>search_leads</code> with a country filter), run as the researcher account of Łukasz Abramek (user 12, role RESEARCHER)

For every assistant run we save the exact prompt, the full answer text, a full-page screenshot, a screenshot of the connector menu (as evidence of the on/off state), the model name shown in the UI, and the response time. The files are in `raw/`. **Copilot is Laser Components' own platform and was not run.**

Timing. Ground-truth queries were run on 2026-09-29 (before 15:15 CEST) against the live server at `127.0.0.1:3003`. Assistant runs: 2026-09-29, 15:18–15:55 CEST (the two *Claude + LC* runs in Sections 3.1/4.1 are the 15:44–15:49 CEST batch), plus the 22:18 CEST *Claude + LC* Poland re-verification (3.4), the 22:42 CEST *Claude + LC* Czech Republic rerun that stalled (4.4) and its successful 23:02 CEST repeat (4.5), and the 22:43–22:46 CEST *ChatGPT + LC* reruns used in Sections 3.1/4.1.

Run conditions, as found (not changed).

- *Claude.ai*: model label "Opus 5.5 · Medium". Web search was off; Memory was on. Claude Docs and Vendo Connect were switched on in every chat. The LC Connect toggle was off for the *Claude* runs and on for the *Claude + LC* runs. The toggle is remembered between chats, so after the runs it was set back to off. The account's standing "Vendo only" instruction applied to the first four *Claude* runs (15:18–15:31 CEST); it was then lifted and the two *Claude + LC* runs were repeated at 15:44 and 15:48 CEST with the connector actually called (`search_leads` , "Always allow" granted once).
- *ChatGPT*: Free plan, "Myślenie" (Thinking) mode. For the *ChatGPT*(without) runs (15:28–15:31 CEST) no app was selected and LC Connect was not yet installed in the account; ChatGPT ran a web search on its own and cited sources. For the *ChatGPT + LC* runs used here (22:43 and 22:45 CEST) the "LC Connector" app was selected as a chip in the composer before sending. Before these reruns the account owner replaced his ChatGPT custom instructions, which had mentioned Vendo and made the 15:51–15:55 CEST runs call the source "Łącznik Vendo" (Appendix A).

ChatGPT baseline (why the "without" columns are the afternoon runs). The *ChatGPT*(without) runs were also repeated at 22:40–22:42 CEST with no app selected (`raw/chatgpt-without-Q1-v2.md` , `-Q2-v2.md`). They are **invalid as a baseline**: with LC Connect installed in the account and the new instruction in place, ChatGPT invoked LC Connect on its own although no app was selected — both answers open "I checked LC Connect first" / "Using LC Connect first", and the server log shows 23 and 15 tool calls respectively. The *ChatGPT*(without) baseline therefore stays the 15:28–15:31 CEST runs (`raw/chatgpt-without-Q1.md` , `-Q2.md` : Poland 6 named / 4 in base; Czech Republic 5 named / 0 in base), made before the app existed in the account.

Practical note for the customer: to show a true "without" in ChatGPT, use a temporary chat or disconnect the app — an installed app may be used automatically even when it is not selected. - **Screenshots.** They show the visible browser window, not the full scrolled page. The complete answer text is in `raw/<product>-<with|without>-<Q1|Q2>.md` (the repeated *Claude + LC* runs are `raw/claude-with-Q1-v2.md` / `raw/claude-with-Q2-v2.md` ; the *ChatGPT + LC* reruns are `raw/chatgpt-with-Q1-v2.md` / `raw/chatgpt-with-Q2-v2.md`). -

Database changed during the runs. At 15:19:50 CEST another work stream ran a country back-fill (commit `d280649` , "26 PL/CZ leads assigned, tagged geo:text, revertable"). Poland now has **25** country-tagged leads and the Czech Republic **29**. All assistant runs used in this document postdate this change; the database lists and overlap figures are a live pull taken at 15:39 CEST. - **What the back-fill did.** The back-fill (`scripts/backfill-geography-text.ts`) is evidence-based: it assigned a country to a lead only where the lead's own description text names a country or a city, and it tagged every changed row `geo:text` so the change is identifiable and revertible. This is not GPS or otherwise independently verified location data — it reflects what the research notes say about a company, not a confirmed address — so it should be read as directional evidence, not ground truth.

Server changes made during and after the evaluation.

- **15:35 CEST (before the "+ LC" runs used in this document):** the `search_leads` and `get_leads` MCP tools were changed to accept a combined country + product filter in the same call, and to show a country column in the results. All "+ LC" runs used in this document postdate this change. The superseded 15:51–15:55 CEST ChatGPT + LC runs still needed 30 tool calls per question (Appendix A); the 22:43–22:46 CEST reruns needed 22 (Poland) and 2 (Czech Republic).
- **Later today, after the runs:** the model-visible `search_leads` / `get_leads` result was made more compact (fewer/shorter fields per row), and a presentation rule was added asking the connected assistant to answer per-item fields (city, product, confidence) in text rather than leaving the reader to open the widget. Both changes were made *after* the 15:18–15:55 CEST runs and are not reflected in them — see the "widget only, no per-company text" finding for both *Claude + LC* runs in Sections 3.3, 4.3 and 5. The 22:18 CEST Claude re-verification (3.4), the 22:42 CEST Czech rerun that stalled (4.4), its 23:02 CEST successful repeat (4.5) and the 22:43–22:46 CEST ChatGPT + LC reruns postdate both changes.
- **Enrichment was not available all day.** Every `enrich_lead` call on 2026-09-29 failed because the research API key (`PERPLEXITY_API_KEY`) was not configured on the server, so web enrichment played no part in any answer compared here. The research service was configured and verified later the same night (23:1x CEST, after all runs compared in this document) — see `raw/enrich-verified.txt`.

2. What the LC Connect database contains

This section reflects a live database pull at 2026-09-29 15:39 CEST, superseding the 15:00 snapshot used in the first draft.

- **396 leads in total.** 177 have a country assigned. **219 have no country** and no location text.
- **Pulsed laser diode leads can be identified.** Every lead is linked to exactly one Laser Components product, meaning the product it is a prospect for. We define "PLD-linked" as a product name matching `Pulsed Laser Diodes%`, which covers *Pulsed Laser Diodes at 905 nm*, *Pulsed Laser Diodes at 1550 nm* and (one Polish lead) *Pulsed Laser Diodes with Fiber Pigtail*. *High Power Laser Diodes* (one Polish lead) does not match that filter and is not counted as PLD-linked — see 3.2. The link records a researched **sales fit**; it does not verify that the company buys PLDs today. The lead descriptions say so plainly, with wording such as "Needs pulsed laser diodes", "Potential buyer" and "HIGH/MEDIUM/LOW FIT".
- **City.** The structured location field is empty for all of these leads. Where a city is known, it appears only inside the free-text description. We quote it below and mark it (*desc.*); where the description gives no city, the field is left blank rather than guessed.
- **Status and owner.** All Polish and Czech leads have status NEW and no owner assigned. None has a confidence score.
- **How the connector reaches them.** `search_leads` filters by country but not by product. `get_leads` filters by product but not by country. An assistant therefore had to filter by country and read the product column separately at the time of the runs (15:18–15:31 CEST) — see the note at the end of this section about a tool change made afterward.

Data-quality finding: leads without a country (partially resolved)

The first draft flagged that a country filter misses leads that are clearly Polish or Czech from their legal form or well-known identity but had no `country_id` set. **This has now been partly fixed** by the evidence-based back-fill described in Section 1 — CRW Telesystem-Mesko, TopGaN, Fluence Technology, FIBRAIN, Opt Lasers, VIGO Photonics, PIT-RADWAR, Solaris Optics and SDS Optic all now carry `country_id = Poland`, and Meopta

and CRYTUR now carry `country_id = Czech Republic`. **219 leads (of 396) still have no country set**, and a country filter still under-reports until those are resolved. **Recommendation stands:** continue filling in missing countries.

3. Q1: Poland

3.1 Comparison table

LC database column refreshed 2026-09-29 15:39 CEST (live pull, post-backfill), replacing the 14-lead pre-backfill snapshot used in the first draft. The **Claude + LC** column is the repeated, connector-invoked run (15:44 CEST; for the post-fix run see 3.4) and the **ChatGPT + LC** column is the 22:43 CEST rerun (`raw/chatgpt-with-Q1-v2.md`); the first *Claude + LC* attempt, where the connector was on but never called, is superseded — see the note below the table. Matched companies (present in both an assistant's answer and the LC database) are **bold**.

CLAUDE	CLAUDE + LC †	CHATGPT	CHATGPT + LC ‡	LC DATABASE (25)
PCO S.A.	Semicon Sp. z o.o.	CRW Telesystem-Mesko	Semicon Sp. z o.o.	Autocomp Management Sp. z o.o.
CRW Telesystem-Mesko	LiDAR3D Sp. z o.o.	TOYA S.A. / YATO	WITU	CRW Telesystem-Mesko Sp. z o.o.
Mesko S.A.	Instytut Optoelektroniki WAT	AiQ robotics P.S.A.	Perspectiva Solutions	CTL Laserinstruments Sp. z o.o.
Dezamet S.A.	Wojskowy Instytut Techniczny Uzbrojenia (WITU)	MESKO S.A.	CTL Laserinstruments	Etronika Sp. z o.o.
WB Group (WB Electronics)	SDS Optic S.A.	PCO S.A.	OEM Tech	FIBRAIN Sp. z o.o.
PIT-RADWAR	Perspectiva Solutions	WB Electronics / WB Group	Zakłady Mechaniczne Tarnów (ZMT)	Fluence Technology Sp. z o.o.
Huta Stalowa Wola	Lukasiewicz-PIAP		Autocomp Management	Huta Stalowa Wola S.A. (HSW)
Advanced Protection Systems	CTL Laserinstruments Sp. z o.o.		Mesko S.A.	Inframet
Delta Optical	Solaris Optics S.A.		Instytut Optoelektroniki WAT	Instytut Optoelektroniki WAT
Łukasiewicz – Institute of Microelectronics and Photonics	Opt Lasers (Tomorrow's System Sp. z o.o.)		Huta Stalowa Wola (HSW)	LiDAR3D Sp. z o.o.
WAT, Institute of Optoelectronics	OEM Tech Sp. z o.o.		WB Electronics / WB Group	Lukasiewicz-PIAP
VIGO Photonics	Huta Stalowa Wola S.A. (HSW)		Etronika	Mesko S.A.
	Zakłady Mechaniczne Tarnów S.A. (ZMT)		PIT-RADWAR	OEM Tech Sp. z o.o.
	Autocomp Management Sp. z o.o.		PCO S.A.	Opt Lasers (Tomorrow's System Sp. z o.o.)

CLAUDE	CLAUDE + LC †	CHATGPT	CHATGPT + LC ‡	LC DATABASE (25)
	(+ 11 more rows below the widget's scroll — full set = all 25 rows in the LC database column)		CRW Telesystem-Mesko	PCO S.A.
			LiDAR3D	Perspectiva Solutions
				PIT-RADWAR S.A.
				SDS Optic S.A.
				Semicon Sp. z o.o.
				Solaris Optics S.A.
				TopGaN Sp. z o.o.
				VIGO Photonics S.A.
				WB Electronics S.A. (WB Group)
				Wojskowy Instytut Techniczny Uzbrojenia (WITU)
				Zakłady Mechaniczne Tarnów S.A. (ZMT)

Notes on the "+ LC" columns (apply throughout this document, Poland and Czech Republic alike): † Claude + LC — the connector was called this time (`search_leads` with a country filter only), and returned the *entire* country list as the LC Connect "Leads — search" widget (25 of 25 rows for Poland, 29 of 29 for the Czech Republic). Claude wrote no text answer naming individual companies, did not filter to pulsed-laser-diode users, and gave no city, product/application or confidence per company — it pointed to the widget and offered to re-run the search or pull per-lead detail. Because the answer is the unfiltered database list, **the match against the LC database is the full column by construction**, not a subset Claude selected; the table shows only the 14 (of 25) and 14 (of 29) rows visible in the saved screenshots before the widget's inner scroll cut them off — the remaining rows are the same list, continued, in the LC database column. The first attempt at this run (superseded, not shown in this table) found the connector switched on but never called, because of an account-level standing instruction restricting connector use to Vendo; that instruction was lifted and the run above is the repeat. ‡ ChatGPT + LC (22:43–22:46 CEST reruns, after the custom-instruction change) — the app is named "LC Connector" in ChatGPT's UI. Poland: 22 `search_leads` calls (20 × query+country, 2 × country+productName), all "ok", no `enrich_lead`, no web search, answered in 9 s (UI) — in Polish, although the prompt was English. It named 16 companies (its intro says 15), all in the base: 12 PLD-linked, plus 4 LiDAR-linked prospects (PIT-RADWAR, PCO, CRW Telesystem-Mesko, LiDAR3D) with wavelength "nieokreślone" (unspecified). It marked cities not stored in LC ("brak w LC") and starred the ones it added from public knowledge. Czech Republic: 2 `search_leads` calls (country+productName), a few seconds, 14 named, all 14 PLD-linked; city "Not in LC Connect" and confidence "Not recorded" instead of guessing. Both answers name the source "LC Connect". The earlier 15:51–15:55 CEST runs are in Appendix A.

ChatGPT also mentioned Bumar Soldier S.A. as an R&D lead, but said it would not count it as a PLD user (its 1535 nm source is a solid-state microchip laser). Bumar Soldier is not in the LC database.

3.2 LC database detail (Poland, country filter)

Live pull, 2026-09-29 15:39 CEST (25 leads with `country_id = Poland`; supersedes the 14-lead 15:00 snapshot). 16 of the 25 leads are linked to a product name matching `Pulsed Laser Diodes%` (10 × 905 nm, 5 × 1550 nm, 1 × fiber-pigtailed). The other 9 are linked to LiDAR (7), High Power Laser Diodes (1) or Silicon-PIN

photodiodes (1) — the last two do not match the **Pulsed Laser Diodes%** filter and are outside the PLD-linked subset in (b) below. City is quoted only where the lead's own description names one, marked *(desc.)*; otherwise it is left blank.

(a) All 25 Poland leads

LEAD	CITY	LC PRODUCT LINK	WEBSITE	STATUS	OWNER
Autocomp Management Sp. z o.o.	Szczecin <i>(desc.)</i>	PLD 905 nm	ac-m.pl	NEW	—
CRW Telesystem-Mesko Sp. z o.o.	—	LiDAR	telesystem.eu	NEW	—
CTL Laserinstruments Sp. z o.o.	Warsaw <i>(desc.)</i>	PLD 905 nm	ctl.com.pl	NEW	—
Etronika Sp. z o.o.	—	PLD 1550 nm	etronika.pl	NEW	—
FIBRAIN Sp. z o.o.	—	PLD, fiber-pigtailed	fibrain.com	NEW	—
Fluence Technology Sp. z o.o.	Warsaw <i>(desc.)</i>	PLD 905 nm	fluence.technology	NEW	—
Huta Stalowa Wola S.A. (HSW)	Stalowa Wola <i>(desc.)</i>	PLD 1550 nm	hsw.pl	NEW	—
Inframet	—	LiDAR	inframet.com	NEW	—
Instytut Optoelektroniki WAT	Warsaw <i>(desc.)</i>	PLD 1550 nm	ioe.wat.edu.pl	NEW	—
LiDAR3D Sp. z o.o.	—	LiDAR	lidar3d.pl	NEW	—
Lukasiewicz-PIAP	Warsaw <i>(desc.)</i>	LiDAR	piap.lukasiewicz.gov.pl	NEW	—
Mesko S.A.	Skarżysko-Kamienna <i>(desc.)</i>	PLD 905 nm	mesko.com.pl	NEW	—
OEM Tech Sp. z o.o.	—	PLD 905 nm	oem-tech.pl	NEW	—
Opt Lasers (Tomorrow's System Sp. z o.o.)	Warsaw <i>(desc.)</i>	High Power Laser Diodes	optlasers.com	NEW	—
PCO S.A.	—	LiDAR	pcosa.com.pl	NEW	—
Perspectiva Solutions	—	PLD 905 nm	perspectivasolutions.com	NEW	—
PIT-RADWAR S.A.	—	LiDAR	en.pitradwar.com	NEW	—
SDS Optic S.A.	Lublin <i>(desc.)</i>	Silicon-PIN photodiodes	sdsoptic.com	NEW	—
Semicon Sp. z o.o.	Warszawa <i>(desc.)</i>	PLD 905 nm	semicon.com.pl	NEW	—
Solaris Optics S.A.	near Warsaw <i>(desc.)</i>	PLD 1550 nm	solarisoptics.eu	NEW	—
TopGaN Sp. z o.o.	—	PLD 905 nm	topganlasers.com	NEW	—
VIGO Photonics S.A.	—	LiDAR	vigophotonics.com	NEW	—
WB Electronics S.A. (WB Group)	—	PLD 1550 nm	wbgroup.pl	NEW	—
Wojskowy Instytut Techniczny Uzbrojenia (WITU)	Zielonka <i>(desc.)</i>	PLD 905 nm	witu.mil.pl	NEW	—
Zakłady Mechaniczne Tarnów S.A. (ZMT)	Tarnów <i>(desc.)</i>	PLD 905 nm	zmt.tarnow.pl	NEW	—

(b) PLD-linked subset (16 of 25 — product name ILIKE 'Pulsed Laser Diodes%')

LEAD	CITY	LC PRODUCT LINK	WEBSITE	STATUS	OWNER
Autocomp Management Sp. z o.o.	Szczecin <i>(desc.)</i>	PLD 905 nm	ac-m.pl	NEW	—

LEAD	CITY	LC PRODUCT LINK	WEBSITE	STATUS	OWNER
CTL Laserinstruments Sp. z o.o.	Warsaw <i>(desc.)</i>	PLD 905 nm	ctl.com.pl	NEW	—
Etronika Sp. z o.o.	—	PLD 1550 nm	etronika.pl	NEW	—
FIBRAIN Sp. z o.o.	—	PLD, fiber-pigtailed	fibrain.com	NEW	—
Fluence Technology Sp. z o.o.	Warsaw <i>(desc.)</i>	PLD 905 nm	fluence.technology	NEW	—
Huta Stalowa Wola S.A. (HSW)	Stalowa Wola <i>(desc.)</i>	PLD 1550 nm	hsw.pl	NEW	—
Instytut Optoelektroniki WAT	Warsaw <i>(desc.)</i>	PLD 1550 nm	ioe.wat.edu.pl	NEW	—
Mesko S.A.	Skarżysko-Kamienna <i>(desc.)</i>	PLD 905 nm	mesko.com.pl	NEW	—
OEM Tech Sp. z o.o.	—	PLD 905 nm	oem-tech.pl	NEW	—
Perspectiva Solutions	—	PLD 905 nm	perspectivasolutions.com	NEW	—
Semicon Sp. z o.o.	Warszawa <i>(desc.)</i>	PLD 905 nm	semicon.com.pl	NEW	—
Solaris Optics S.A.	near Warsaw <i>(desc.)</i>	PLD 1550 nm	solarisoptics.eu	NEW	—
TopGaN Sp. z o.o.	—	PLD 905 nm	topganlasers.com	NEW	—
WB Electronics S.A. (WB Group)	—	PLD 1550 nm	wbgroup.pl	NEW	—
Wojskowy Instytut Techniczny Uzbrojenia (WITU)	Zielonka <i>(desc.)</i>	PLD 905 nm	witu.mil.pl	NEW	—
Zakłady Mechaniczne Tarnów S.A. (ZMT)	Tarnów <i>(desc.)</i>	PLD 905 nm	zmt.tarnow.pl	NEW	—

All 25 have a website stored. Status NEW, no owner, no confidence score, for all 25.

3.3 Assessment per run (Q1)

Overlap is computed against the current 25-lead Poland list (2026-09-29 15:39 CEST pull), matching company names case-insensitively, stripping legal-form suffixes (Sp. z o.o., S.A., etc.) and normalizing diacritics; "WAT, Institute of Optoelectronics" / "Łukasiewicz – PIAP" are matched to "Instytut Optoelektroniki WAT" / "Łukasiewicz-PIAP" as the same entities under a translated/reordered name.

RUN	COMPANIES NAMED	MATCHED IN LC DATABASE (25)	MATCHED COMPANY NAMES	RESPONSE TIME
Claude	12	8	PCO S.A., CRW Telesystem-Mesko Sp. z o.o., Mesko S.A., WB Electronics S.A. (WB Group), PIT-RADWAR S.A., Huta Stalowa Wola S.A. (HSW), Instytut Optoelektroniki WAT, VIGO Photonics S.A.	~43 s
Claude + LC †	25 (all rows in the widget; text answer names none individually)	25	full 25-lead Poland list, by construction — see 3.2	~54 s (incl. one "Always allow" click)
ChatGPT	6	4	CRW Telesystem-Mesko Sp. z o.o., MESKO S.A. → Mesko S.A., PCO S.A., WB Electronics/WB Group → WB Electronics S.A. (WB Group)	~31 s thinking (UI), done within 48 s

RUN	COMPANIES NAMED	MATCHED IN LC DATABASE (25)	MATCHED COMPANY NAMES	RESPONSE TIME
ChatGPT + LC ‡	16	16 (12 PLD-linked, 4 LiDAR-linked)	Semicon, WITU, Perspektiva Solutions, CTL Laserinstruments, OEM Tech, ZMT, Autocomp Management, Mesko, Instytut Optoelektroniki WAT, HSW, WB Electronics (WB Group), Etronika, PIT-RADWAR, PCO, CRW Telesystem-Mesko, LiDAR3D — all 16 matched, none new; missing from the 16 PLD-linked: Solaris Optics, FIBRAIN, Fluence Technology, TopGaN	9 s (UI), complete within ~37 s wall clock
LC database	25	25	(all 25, see 3.2)	under 1 s (direct query)

Not matched: Claude named Dezamet, Advanced Protection Systems, Delta Optical and Łukasiewicz – Instytut Mikroelektroniki i Fotoniki, none of which are in the base. Claude + LC named none unmatched — its answer is the full database list by construction (25 of 25 matched). ChatGPT named TOYA/YATO and AiQ robotics, neither in the base. ChatGPT + LC named no company outside the base — all 16 it listed are of the 25. These are possible new leads to check, not errors — no website was verified in this test.

3.4 Claude + LC Connect after the server fix (same prompt, 22:18 CEST)

Run: claude.ai, new chat, same Q1 prompt as above, LC Connect on. Sent **2026-09-29 22:17:59 CEST**. Model label (UI): Opus 5.5 · Medium. Response completed in ~**31 s** (excluding the one "Always allow" permission dialog, granted immediately, ~10 s into the run). Chat: <https://claude.ai/chat/a9a3efd2-6cbc-4c67-b228-266dda251994>.

What changed on the server first. Two commits landed after the 15:44 CEST run used in Section 3.1: [d16086e](#) (compact rows in the tool result's `structuredContent`, plus a presentation rule asking the connected assistant to answer per-item fields in text) and [a0dfd53](#) (rows are now *also* rendered as plain text in the tool result — because claude.ai forwards only a tool result's text part to the model, not `structuredContent`; `search_leads`'s free-text query was also made to match product names, not just company names). An intermediate check at **16:03 CEST** ([raw/claude-with-Q1-v3.md](#), after [d16086e](#) but before [a0dfd53](#)) still failed: `search_leads` reported 25 Polish leads, 9 of them matching "rangefinder," and the widget card rendered, but — in Claude's words — "the rows never reached me." Claude could read only 3 of the 25 leads (via `advanced_search` calls that happened to return few enough rows to appear inline) and asked the user to paste a CSV to finish the table.

Tool calls (22:18 run): 1. `search_leads` — `{"country": "Poland", "productName": "Pulsed Laser Diodes", "limit": 100}` → 16 of 16 PLD-linked Polish leads, in one call. 2. `search_leads` — a free-text "Poland" search, to catch leads without a country assigned → 1 of 1 record (Perspektiva Solutions), already among the 16 — found nothing new.

Claude's answer (verbatim table, [raw/claude-with-Q1-v4.md](#)). Claude stated that a product link in LC Connect means "sales fit," not confirmed use, and labelled the city, product/application and confidence columns as its own general knowledge — unverified, since LC Connect stores no city field for these leads. It also flagged, unprompted, that PCO S.A. — Poland's best-known maker of laser rangefinders — is not among the 16 PLD-linked leads.

LEAD (LC CONNECT)	PLD FIT (LC CONNECT)	CITY (UNVERIFIED)	LIKELY PRODUCT / APPLICATION (UNVERIFIED)	CONFIDENCE OF ACTUAL PLD USE (UNVERIFIED)
Instytut Optoelektroniki WAT	1550 nm	Warsaw	R&D on laser rangefinders, LiDAR, laser warning	High

LEAD (LC CONNECT)	PLD FIT (LC CONNECT)	CITY (UNVERIFIED)	LIKELY PRODUCT / APPLICATION (UNVERIFIED)	CONFIDENCE OF ACTUAL PLD USE (UNVERIFIED)
WB Electronics (WB Group)	1550 nm	Ożarów Mazowiecki	Fire control, UAV and loitering munition sensors	Medium
Etronika	1550 nm	Warsaw area	Defence optoelectronics, observation devices	Medium
Mesko S.A.	905 nm	Skarżysko-Kamienna	Ammunition and missiles (proximity fuzes are the plausible fit)	Medium
WITU	905 nm	Zielonka	Weapons and ammunition test and R&D	Medium
Semicon Sp. z o.o.	905 nm	Komorów (near Warsaw)	Laser modules	Medium
Autocomp Management	905 nm	Szczecin	Simulators (fits only if they build laser engagement systems)	Low–medium
Huta Stalowa Wola (HSW)	1550 nm	Stalowa Wola	Vehicles and turrets; rangefinders more likely bought in as subsystems	Low–medium
ZMT Tarnów	905 nm	Tarnów	Weapons systems, possibly fire control	Low–medium
CTL Laserinstruments	905 nm	?	Medical and industrial lasers (905 nm pulsed therapy devices fit)	Low–medium
OEM Tech	905 nm	?	Laser electronics	Low
Perspectiva Solutions	905 nm	?	Photonics manufacturing	Low
Solaris Optics S.A.	1550 nm	Nadarzyn	Optical components, not light sources	Low
FIBRAIN	pigtailed PLD	Zaczernie (Rzeszów)	Fibre optics, possibly OTDR-type test gear	Low
Fluence Technology	905 nm	Warsaw	Femtosecond fibre lasers, so a different technology	Low
TopGaN	905 nm	Warsaw	GaN laser diode maker, more likely a competitor than a user	Low

Compared with the earlier runs. The 15:44 CEST run (Section 3.1, marked †) returned the same kind of widget but no per-company text at all — Claude pointed at the card and offered to pull detail on request. The 16:03 CEST v3 run had text but could read only 3 of 25 rows, because the rows lived only in `structuredContent`, which claude.ai does not forward to the model. The 22:18 CEST run, after both fixes, read and wrote out all 16 PLD-linked rows in two calls total, with a per-company table matching the format the original question asked for.

4. Q2: Czech Republic

4.1 Comparison table

LC database column refreshed 2026-09-29 15:39 CEST (live pull, post-backfill), replacing the 13-lead pre-backfill snapshot used in the first draft. As in Section 3.1, the **Claude + LC** column shown below is the pre-fix connector-invoked run of 15:48 CEST (a post-fix rerun stalled, 4.4; a repeat succeeded at 23:02 CEST, 4.5 — see 4.5 for the per-company result) and the **ChatGPT + LC** column is the 22:45 CEST rerun (`raw/chatgpt-with-`

Q2-v2.md); the first *Claude + LC* attempt (connector on, never called) is superseded. Matched companies are **bold**.

CLAUDE	CLAUDE + LC †	CHATGPT	CHATGPT + LC ‡	LC DATABASE (29)
Meopta – optika	Ray Service a.s.	ARCHER-LPP s.r.o.	Ray Service a.s.	Aero Vodochody Aerospace a.s.
Optokon	Lightigo s.r.o.	SEFOR Solutions s.r.o.	Lightigo s.r.o.	Argotech a.s.
Safibra	LASER-TECH spol. s r.o.	OptiDy nex s.r.o. / Night Pearl	LASER-TECH spol. s r.o.	CRYTUR spol. s r.o.
Valeo (R&D centre, Prague)	Saab Czech Republic (Training Systems)	Abacus Electric / EVOLVEO	Saab Czech Republic (Training Systems)	Elya Solutions s.r.o.
Camea	URC Systems spol. s r.o.	FOMEI s.r.o.	IPG Photonics (Czech Republic) s.r.o.	EVPU Defence a.s.
Prototypa-ZM	IPG Photonics (Czech Republic) s.r.o.		Aero Vodochody Aerospace a.s.	Excalibur Army spol. s r.o.
MESIT (defence division)	Optaglio s.r.o.		Tatra Defence Vehicle a.s.	Fretech Aerospace s.r.o.
Varroc Lighting Systems	Aero Vodochody Aerospace a.s.		Excalibur Army spol. s r.o.	GenEvo s.r.o.
Primoco UAV / Workswell	Tatra Defence Vehicle a.s.		Narran s.r.o.	Glomex Military Supplies s.r.o.
VTÚ (Military Technical Institute)	Excalibur Army spol. s r.o.		Glomex Military Supplies s.r.o.	IPG Photonics (Czech Republic) s.r.o.
	VUTS a.s. Liberec		LPP Holding a.s.	LASER-TECH spol. s r.o.
	Narran s.r.o.		Elya Solutions s.r.o.	Lightigo s.r.o.
	GenEvo s.r.o.		Primoco UAV SE	LPP Holding a.s.
	Micro-Epsilon Czech Republic s.r.o.		VR Group a.s.	Meopta - optika s.r.o.
	(+ 15 more rows below the widget's scroll – full set = all 29 rows in the LC database column)			Micro-Epsilon Czech Republic s.r.o.
				Narran s.r.o.
				Optaglio s.r.o.
				OPTOKON a.s.
				Pramacom-HT s.r.o.
				Primoco UAV SE
				Ray Service a.s.
				RETIA a.s.
				Saab Czech Republic (Training Systems)
				SQS Vlaknova optika a.s.

CLAUDE	CLAUDE + LC †	CHATGPT	CHATGPT + LC ‡	LC DATABASE (29)
				Tatra Defence Vehicle a.s.
				URC Systems spol. s r.o.
				VR Group a.s.
				VUTS a.s. Liberec
				Workswell s.r.o.

ChatGPT also named Astrum LT (Kralupy nad Vltavou) as a possible laser-diode *supplier*, not a user; Astrum LT is not in the LC database. ChatGPT + LC separately noted that the base has no Czech lead tagged 1550 nm PLD (only 905 nm).

4.2 LC database detail (Czech Republic, country filter)

Live pull, 2026-09-29 15:39 CEST (29 leads with `country_id = Czech Republic`; supersedes the 13-lead 15:00 snapshot). 14 of the 29 leads are linked to PLD 905 nm; the other 15 are linked to LiDAR (9) or Silicon-PIN photodiodes (6). No Czech lead is linked to PLD 1550 nm, although one description (Excalibur Army) names 1550 nm PLDs as a potential need. City is quoted only where the lead's own description names one, marked (*desc.*); otherwise it is left blank, including where the city appears only in the company's own name (e.g. VUTS a.s. **Liberec**) rather than in the description text.

(a) All 29 Czech Republic leads

LEAD	CITY	LC PRODUCT LINK	WEBSITE	STATUS	OWNER
Aero Vodochody Aerospace a.s.	—	PLD 905 nm	aero.cz	NEW	—
Argotech a.s.	Náchod (<i>desc.</i>)	LiDAR	argotech.cz	NEW	—
CRYTUR spol. s r.o.	Turnov (<i>desc.</i>)	LiDAR	crytur.com	NEW	—
Elya Solutions s.r.o.	—	PLD 905 nm	elya.cz	NEW	—
EVPU Defence a.s.	—	LiDAR	evpudefence.com	NEW	—
Excalibur Army spol. s r.o.	Šternberk / Pardubice (<i>desc.</i>)	PLD 905 nm	excaliburarmy.cz	NEW	—
Frentech Aerospace s.r.o.	Brno (<i>desc.</i>)	LiDAR	frentech.cz	NEW	—
GenEvo s.r.o.	—	Silicon-PIN photodiodes	genevo.com	NEW	—
Glomex Military Supplies s.r.o.	—	PLD 905 nm	glomex-ms.com	NEW	—
IPG Photonics (Czech Republic) s.r.o.	Prague (<i>desc.</i>)	PLD 905 nm	ipgphotonics.com	NEW	—
LASER-TECH spol. s r.o.	Olomouc (<i>desc.</i>)	PLD 905 nm	laser-tech.cz	NEW	—
Lightigo s.r.o.	—	PLD 905 nm	<i>no website stored</i>	NEW	—
LPP Holding a.s.	—	PLD 905 nm	lpp-holding.com	NEW	—
Meopta - optika s.r.o.	—	LiDAR	meopta.com	NEW	—
Micro-Epsilon Czech Republic s.r.o.	Bechyně (<i>desc.</i>)	Silicon-PIN photodiodes	micro-epsilon.cz	NEW	—
Narran s.r.o.	—	PLD 905 nm	narran.cz	NEW	—

LEAD	CITY	LC PRODUCT LINK	WEBSITE	STATUS	OWNER
Optaglio s.r.o.	Lochovice <i>(desc.)</i>	Silicon-PIN photodiodes	optaglio.com	NEW	—
OPTOKON a.s.	Jihlava <i>(desc.)</i>	LiDAR	optokon.com	NEW	—
Pramacom-HT s.r.o.	—	LiDAR	infrared.cz	NEW	—
Primoco UAV SE	—	PLD 905 nm	primoco.com	NEW	—
Ray Service a.s.	Staré Město <i>(desc.)</i>	PLD 905 nm	rayservice.com	NEW	—
RETIA a.s.	Pardubice <i>(desc.)</i>	LiDAR	retia.eu	NEW	—
Saab Czech Republic (Training Systems)	—	PLD 905 nm	saab.com	NEW	—
SQS Vlaknova optika a.s.	Nová Paka <i>(desc.)</i>	LiDAR	sqs-fiberoptics.com	NEW	—
Tatra Defence Vehicle a.s.	—	PLD 905 nm	tatradv.cz	NEW	—
URC Systems spol. s r.o.	—	Silicon-PIN photodiodes	urc-systems.cz	NEW	—
VR Group a.s.	Prague <i>(desc.)</i>	PLD 905 nm	vrg.cz	NEW	—
VUTS a.s. Liberec	—	Silicon-PIN photodiodes	vuts.cz	NEW	—
Workswell s.r.o.	Prague <i>(desc.)</i>	Silicon-PIN photodiodes	workswell.eu	NEW	—

(b) PLD-linked subset (14 of 29 — product name ILIKE 'Pulsed Laser Diodes%')

LEAD	CITY	LC PRODUCT LINK	WEBSITE	STATUS	OWNER
Aero Vodochody Aerospace a.s.	—	PLD 905 nm	aero.cz	NEW	—
Elya Solutions s.r.o.	—	PLD 905 nm	elya.cz	NEW	—
Excalibur Army spol. s r.o.	Šternberk / Pardubice <i>(desc.)</i>	PLD 905 nm	excaliburarmy.cz	NEW	—
Glomex Military Supplies s.r.o.	—	PLD 905 nm	glomex-ms.com	NEW	—
IPG Photonics (Czech Republic) s.r.o.	Prague <i>(desc.)</i>	PLD 905 nm	ipgphotonics.com	NEW	—
LASER-TECH spol. s r.o.	Olomouc <i>(desc.)</i>	PLD 905 nm	laser-tech.cz	NEW	—
Lightigo s.r.o.	—	PLD 905 nm	<i>no website stored</i>	NEW	—
LPP Holding a.s.	—	PLD 905 nm	lpp-holding.com	NEW	—
Narran s.r.o.	—	PLD 905 nm	narran.cz	NEW	—
Primoco UAV SE	—	PLD 905 nm	primoco.com	NEW	—
Ray Service a.s.	Staré Město <i>(desc.)</i>	PLD 905 nm	rayservice.com	NEW	—
Saab Czech Republic (Training Systems)	—	PLD 905 nm	saab.com	NEW	—
Tatra Defence Vehicle a.s.	—	PLD 905 nm	tatradv.cz	NEW	—
VR Group a.s.	Prague <i>(desc.)</i>	PLD 905 nm	vrg.cz	NEW	—

28 of 29 have a website stored (Lightigo s.r.o. does not). Status NEW, no owner, no confidence score beyond the HIGH/MEDIUM/LOW FIT wording in the descriptions, for all 29.

4.3 Assessment per run (Q2)

Overlap is computed against the current 29-lead Czech Republic list (2026-09-29 15:39 CEST pull), with the same name-normalization rule as 3.3. Claude's "Primoco UAV / Workswell" row names two distinct companies that are two distinct leads in the database, so it is counted as two matches, not one.

RUN	COMPANIES NAMED	MATCHED IN LC DATABASE (29)	MATCHED COMPANY NAMES	RESPONSE TIME
Claude	10 (11 counting Primoco and Workswell separately)	4	Meopta - optika s.r.o., OPTOKON a.s., Primoco UAV SE, Workswell s.r.o.	~38 s
Claude + LC †	29 (all rows in the widget; text answer names none individually)	29	full 29-lead Czech Republic list, by construction — see 4.2	~30 s (no permission dialog; reused Q1's "Always allow")
ChatGPT	5	0	—	~27 s thinking (UI), 38 s to completion
ChatGPT + LC ‡	14	14 (all 14 PLD-linked)	exactly the PLD-linked subset in 4.2(b): Ray Service, Lightigo, LASER-TECH, Saab Czech Republic (Training Systems), IPG Photonics (Czech Republic), Aero Vodochody Aerospace, Tatra Defence Vehicle, Excalibur Army, Narran, Glomex Military Supplies, LPP Holding, Elya Solutions, Primoco UAV, VR Group — all 14 matched, none new	a few seconds (UI), complete within ~36 s wall clock
LC database	29	29	(all 29, see 4.2)	under 1 s (direct query)

Not matched: Claude named Safibra, Valeo (Prague), Camea, Prototypa-ZM, MESIT, Varroc Lighting Systems and VTÚ, none of which are in the base. Claude + LC named none unmatched — its answer is the full database list by construction (29 of 29 matched). ChatGPT named ARCHER-LPP, SEFOR Solutions, OptiDynex/Night Pearl, Abacus Electric/EVOLVEO and FOMEI — none matched; the base separately has *LPP Holding a.s.*, but whether it is related to ARCHER-LPP was not checked, so it is not counted as a match. ChatGPT + LC named no company outside the base — all 14 it listed are of the 29.

4.4 Claude + LC Connect after the server fix, Czech Republic (22:42 CEST) — stalled run, resolved

Run: claude.ai, new chat, same Q2 prompt, LC Connect on ([raw/claude-with-Q2-v4-connectors.jpg](#)). Sent **2026-09-29 22:42:50 CEST**; model label Opus 5.5 · Medium. Chat: <https://claude.ai/chat/c25f0e9c-d7f1-4975-a772-cc05d2466e49>.

Outcome: no answer rendered. The server served every call: 1 combined [search_leads](#) (country + productName) right after sending, then a burst of ~17 free-text [query+country](#) lookups (apparently one per company), then 2 more combined calls at 22:45:17–18 CEST — all outcome "ok". After that the server logged no further activity, but the claude.ai UI still showed "Checking for a response to your message" after ~14 minutes and three reloads ([raw/claude-with-Q2-v4.jpg](#)). Response time could not be measured. (The log window overlaps a ChatGPT session on the same account, so the per-client split is approximate; see [raw/claude-with-Q2-v4.md](#) .)

Root cause and fix. The working hypothesis was the per-company lookup loop, triggered because the Czech leads have empty city fields and Claude tried to fill them one lead at a time. Commit `fbfd16d` (22:58 CEST) shipped both fixes: a one-call rule in the tool guidance against per-company loops, and a city backfill from the lead descriptions (`scripts/backfill-city-text.ts` , tag `geo:city-text` ; 177 leads got a city this way). This closes the open observation — see 4.5 for the successful rerun.

4.5 Claude + LC Connect, Czech Republic, re-verified after the fix (23:02 CEST)

Run: claude.ai, new chat, same Q2 prompt, LC Connect on. Sent **2026-09-29 23:02:09 CEST**; model label Opus 5.5 · Medium. Response completed in **~90 s** (23:02:09 → ~23:03:40 CEST). No permission dialog (the "Always allow" grant from earlier runs the same day still applied). Chat: <https://claude.ai/chat/decbb779-2c28-4fac-b2ef-d334d625f3cd>. Raw: `raw/claude-with-Q2-v5.md` .

Tool calls — the one-call rule held. Two calls total, no per-company loop: 1. `search_leads` — combined country + product filter ("Czech Republic" + a PLD/LiDAR product filter). 2. `search_leads` — free-text "Czech" search, to also catch leads with no country assigned → 30 of 30 records.

Claude's answer. A full per-company text table, split into two sections: 15 leads linked to "Pulsed Laser Diodes at 905 nm" and 9 leads linked to "LiDAR" (24 leads total; six more Czech leads linked only to Silicon-PIN photodiodes were left out). As in the Poland runs, Claude stated that a product link means "sales fit," not confirmed use, and marked city and confidence reasoning as its own unverified general knowledge. It flagged three data gaps unprompted: one matched lead (Metrodat s.r.o.) has no country set in LC Connect, so it was found only by the free-text search, not the country filter; the LiDAR leads carry LiDAR as the product field rather than as an application; and no Czech lead covers 1550 nm PLD or proximity fuzes. Claude's own summary line said "22" Czech leads — a minor miscount against its own 15+9 = 24 rows; this document counts 24.

Compared with the stalled run (4.4). Same prompt, same account state, twelve minutes later in wall-clock terms but after the one-call rule and city backfill landed: two calls instead of a ~20-call burst, and a complete answer in ~90 s instead of a 14-minute stall with nothing rendered. This section is the citable, working Czech Republic **Claude + LC** result; Sections 4.1/4.3 above still show the earlier pre-fix 15:48 CEST widget-only run (kept for the historical record, not re-tabulated here) with a pointer to this section.

5. What this shows

Measured results only:

- **Without the connector, the assistants and the database name mostly different companies.** Poland: Claude named 12 companies (8 matched). ChatGPT named 6 (4 matched). Czech Republic: Claude named 10 (4 matched, counting Primoco and Workswell as two). ChatGPT named 5 (0 matched).
- **With the connector, both assistants returned only companies present in the base.** Claude + LC (15:44/15:48 CEST) returned the entire country list both times (25 of 25 Poland, 29 of 29 Czech Republic) — a 100% match by construction, since it did not filter or select at all; after the fix it returned exactly the 16 Polish PLD-linked leads (3.4). ChatGPT + LC (22:43–22:46 CEST) named 16 Polish leads (12 PLD-linked, 4 LiDAR-linked) and exactly the 14 Czech PLD-linked leads — 30 names, every one in the base, zero outside it.
- **Claude's card-only defect was fixed and re-verified the same day.** The 15:44 CEST runs in this document predate the fix and show it (widget only, no per-company text). After two server fixes (commits `d16086e` , `a0dfd53`), a same-day re-run of the Poland question at 22:17:59 CEST produced a full per-company text table for all 16 PLD-linked Polish leads (Section 3.4).

- **With the combined country + product filter, a single call answers the question.** The 22:18 CEST Claude re-run used `search_leads` with `country` and `productName` together and got the 16 PLD-linked Polish leads in one call (plus one free-text check that found nothing new); ChatGPT + LC answered the Czech question with 2 such calls. The Poland ChatGPT rerun still made 22 `search_leads` calls (mostly free-text per term); the first Claude Czech rerun (4.4) stalled in a per-company lookup pattern, but the one-call rule and city backfill fixed it — the repeat (4.5) took 2 calls and ~90 s.
- **Enrichment played no part.** Every `enrich_lead` call on 2026-09-29 failed because the research API key was not configured on the server; none of the answers in Sections 3–4 used enrichment.
- **Naming.** In the valid reruns ChatGPT names the source "LC Connect". The "Łącznik Vendo" wording of the superseded runs came from the owner's old custom instructions (Appendix A).
- **The overlap is concentrated in large defence firms when there is no connector; with the connector it is broad.** Without LC Connect, PCO, Mesko and WB appear in every Polish answer; almost none of the 29 Czech leads were named. With LC Connect, ChatGPT + LC covered 16 of 25 Polish and 14 of 29 Czech leads, including smaller names (Elya Solutions, Ray Service, Lightigo, LASER-TECH, Narran, Glomex, OEM Tech, LiDAR3D) that neither unconnected assistant guessed.
- **Each unconnected assistant also named companies that are not in the base.** Poland: Claude named 4 unmatched companies, ChatGPT named 2. Czech Republic: Claude named 7 unmatched, ChatGPT named 5. These are possible new leads for the researchers to check, not errors — no website was checked in this test. Neither connected run (" + LC") suggested any company outside the base.
- **How each assistant qualified its answer.** Claude (without the connector), working from memory without web search, gave a confidence level on every row and warned that a rangefinder does not necessarily use a PLD. ChatGPT (without) searched the web without being asked and cited a source for each claim. ChatGPT + LC (Poland) gave a per-company confidence word (*wysoka/średnia*) and marked which cities came from outside LC; ChatGPT + LC (Czech Republic) wrote "Not recorded" for confidence and "Not in LC Connect" for city rather than inferring them.
- **Speed.** Every completed run finished in under a minute and a half — the unconnected assistants in 30–50 s, Claude + LC in 30–54 s, ChatGPT + LC in 9 s / a few seconds of processing (UI), the 23:02 CEST Czech re-verification (4.5) in ~90 s. The exception is the first Czech rerun (4.4), which stalled and never rendered an answer. A direct database query takes under 1 s.
- **Database-side findings from the first draft still hold, with one update.** The missing-country gap has been partly fixed by an evidence-based back-fill (`scripts/backfill-geography-text.ts` , tag `geo:text`) that assigned a country to leads whose own description names one; 26 PL/CZ leads were assigned a country this way, and 219 of 396 leads still have none.

Appendix A. Superseded ChatGPT runs

ChatGPT + LC, 15:51–15:55 CEST (`raw/chatgpt-with-Q1.md` , `raw/chatgpt-with-Q2.md`) — superseded by the 22:43–22:46 CEST reruns. Made while the owner's ChatGPT custom instructions still mentioned Vendo: ChatGPT called the data source "Łącznik Vendo" throughout, called the connector 30 times per question (`search_leads` , `get_leads` , `enrich_lead`) and searched the web unprompted for Q1. All 21 `enrich_lead` calls (8 in Q1, 13 in Q2) failed ("PERPLEXITY_API_KEY environment variable is not set"); ChatGPT reported the failure in the Czech answer. Results: Poland 9 named, 9 in base (Etronika, PCO, Instytut Optoelektroniki WAT, ZMT, Mesko, HSW, WB Electronics, WITU, Semicon); Czech Republic 14 named, 14 in base (EVPU Defence, Meopta, Argotech, Primoco UAV, Elya Solutions, Aero Vodochody, Tatra Defence Vehicle, Excalibur Army, Saab Czech Republic, Ray Service, Lightigo, LASER-TECH, SQS Vlaknova optika, Frentech Aerospace — 9 PLD-linked, 5 LiDAR-linked). Response ~37 s and ~32 s (UI).

ChatGPT "without", 22:40–22:42 CEST (`raw/chatgpt-without-Q1-v2.md` , `raw/chatgpt-without-Q2-v2.md`) — invalid as a baseline. No app was selected, but ChatGPT used the installed LC Connect on its own (Poland: 23

calls, 15 companies named, all in base; Czech Republic: 15 calls incl. 14 failed **enrich_lead**, 14 named, all in base) and supplemented cities from web search. Not used in any comparison; see "ChatGPT baseline" in Section 1.

6. Screenshots

Visible-window captures, 2026-09-29. The complete answer text for each run is in **raw/*.md**.

Claude

New

Projects

Artifacts

Scheduled

Customize

More

Pinned

fastmcp

Chats and tasks

System one I system two w AI

Utworzenie zadania testowego w

Rachunek wyników IKrystyna

Use

Rozbicie pliku rotor-e2e2-202009

Technologia produkcji detali kone

Technologia produkcji detali z koni

Import i analiza rotora z pliku STEP

Import i analiza rotora z pliku STEP

Importowanie rysunku rotora

Importowanie rysunku rotora

Zadania i zdarzenia z tego tygodnia

Oferta dla CFI Wojciech Czarnecki

Bogdan

Mac

Polish companies using pulsed laser diodes

Share

List companies in Poland that use pulsed laser diodes (PLD, 905 nm or 1550 nm) in their products, for example laser rangefinders, LIDAR or proximity fuzes. For each company give: name, city, the product or application, and how confident you are.

Expanding the search to research institutes and radar integrators.

I can't browse right now, so this comes from my training knowledge (through mid-2026). Companies rarely publish which emitter technology they use, so the PLD-specific part is the weakest link in almost every row below. There's also an important distinction to keep in mind. Many military rangefinders use solid-state lasers (1064 nm Nd:YAG or 1535 nm Er:glass) rather than pulsed laser diodes. A company that makes a rangefinder is therefore not automatically a PLD user.

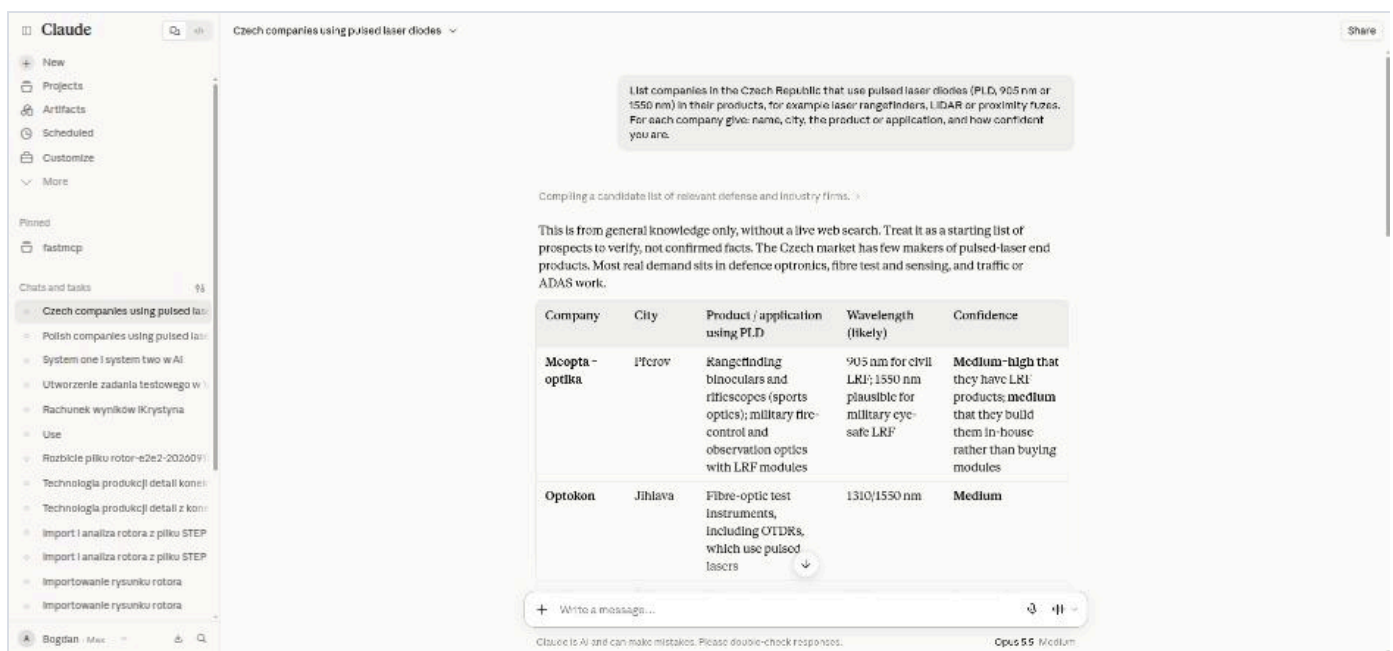
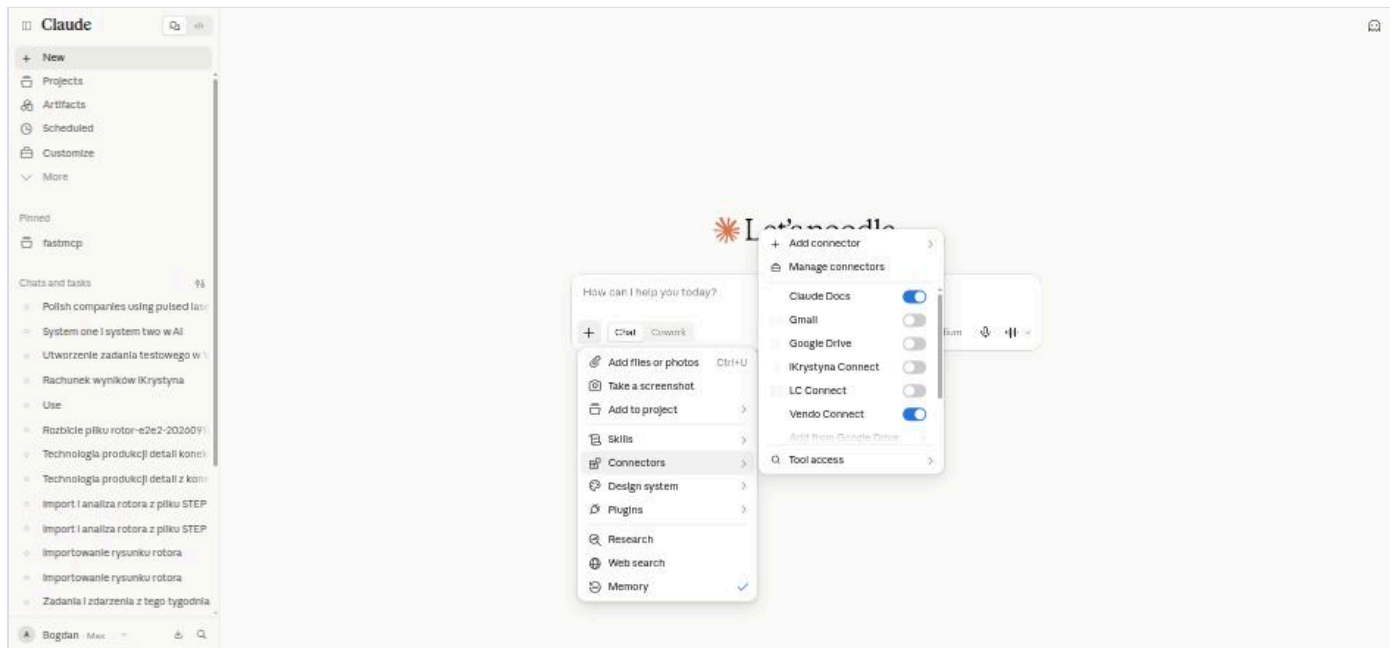
Confidence scale: High = I'm fairly sure they make the product. Medium = product is likely but the PLD use is inferred. Low = plausible lead only.

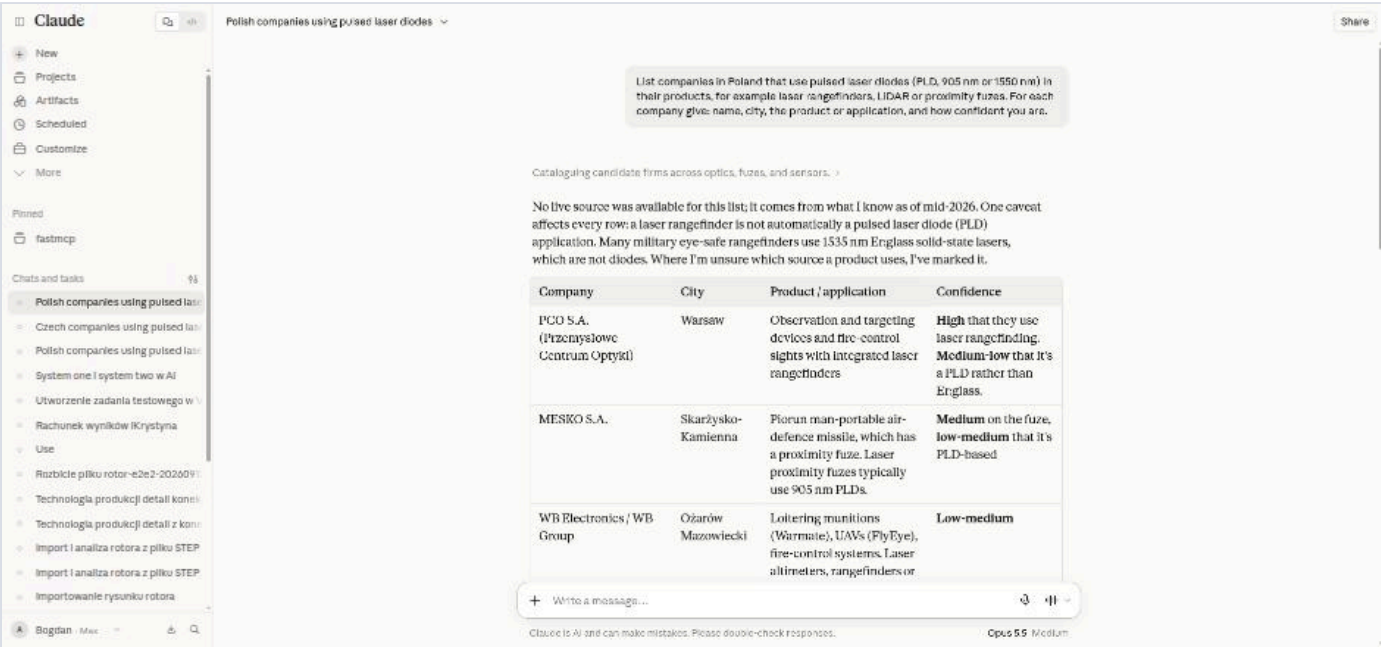
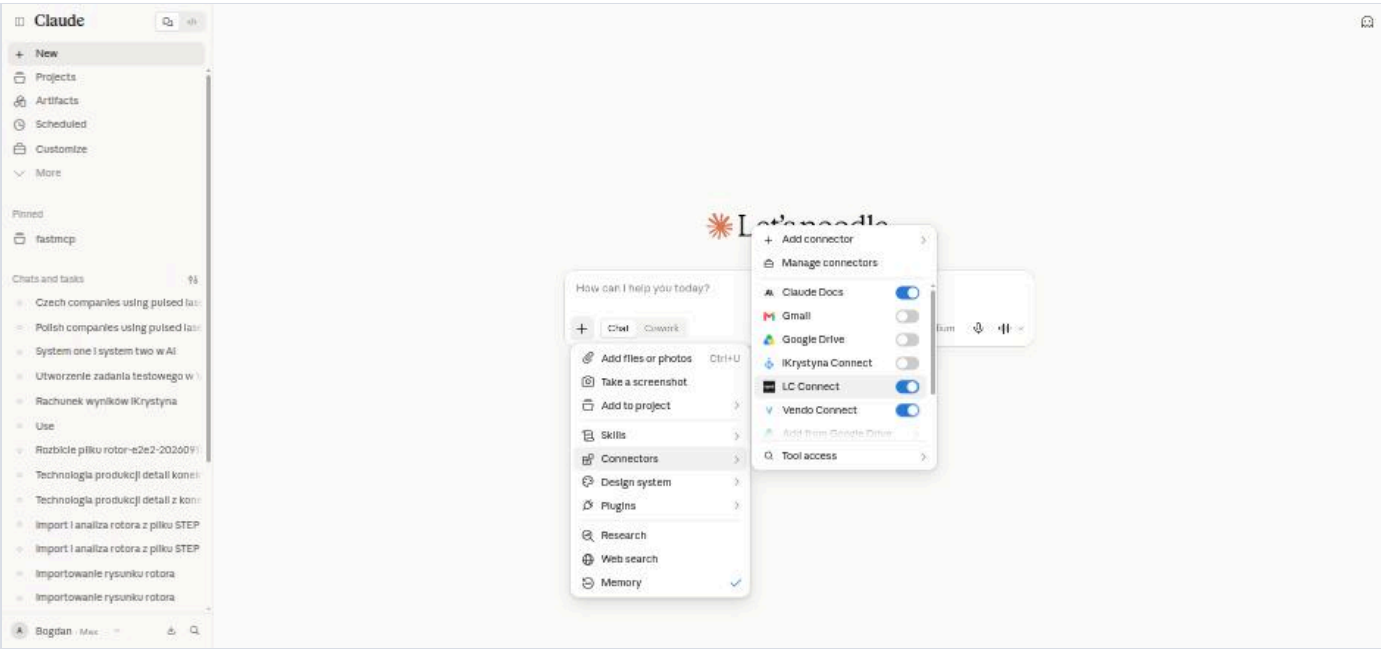
Company	City	Product / application	Confidence
PCO S.A.	Warsaw	Laser rangefinders for fire-control systems, observation devices and turret sights (e.g. ZSSW-30), laser warning systems	High for rangefinders; Medium for PLD (some may be solid-state)
CRW Telesystem-Mesko	Warsaw	Electronics for the Grom/Flotun MANPADS, including the proximity fuze	Medium for fuze work; Low-Medium that the fuze is laser-based

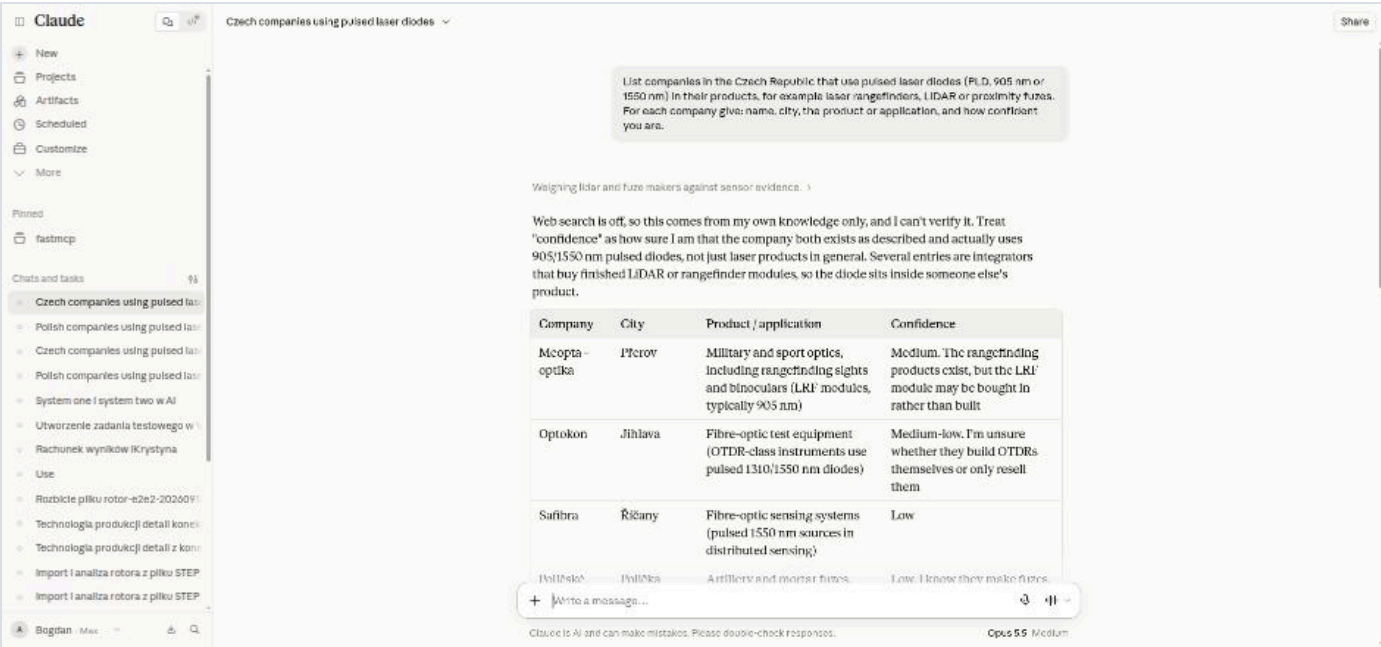
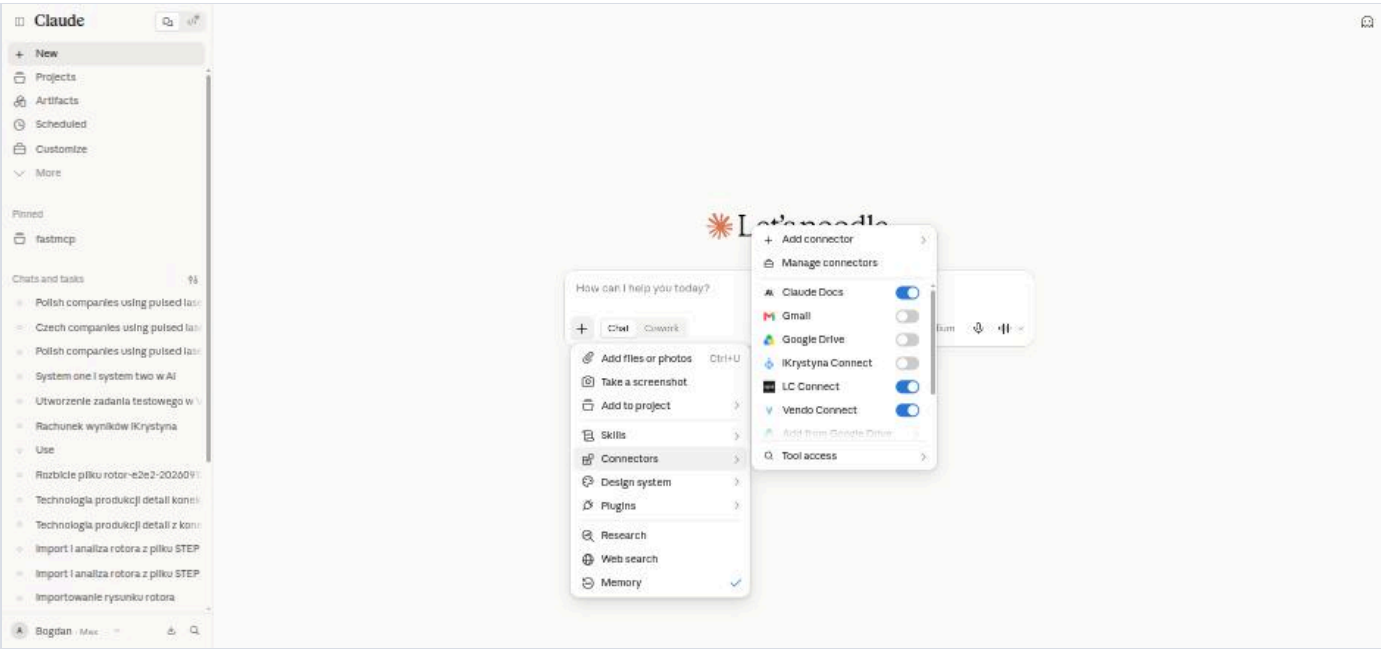
Write a message...

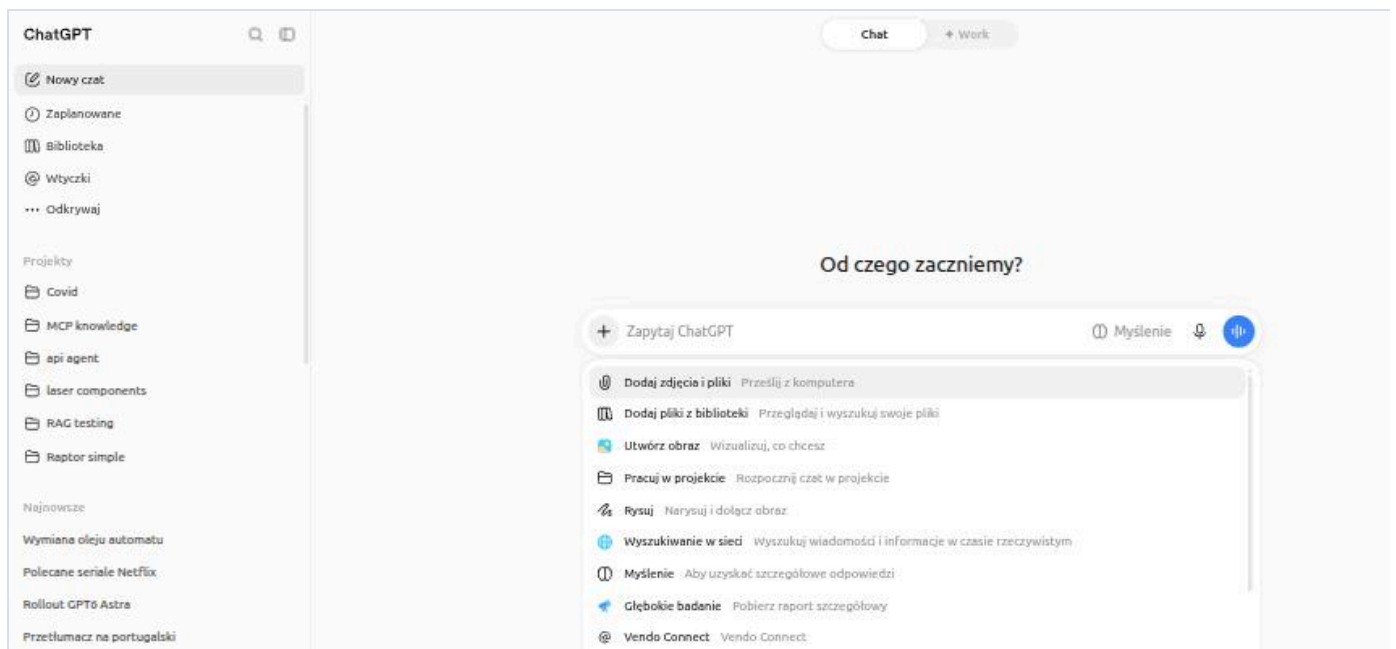
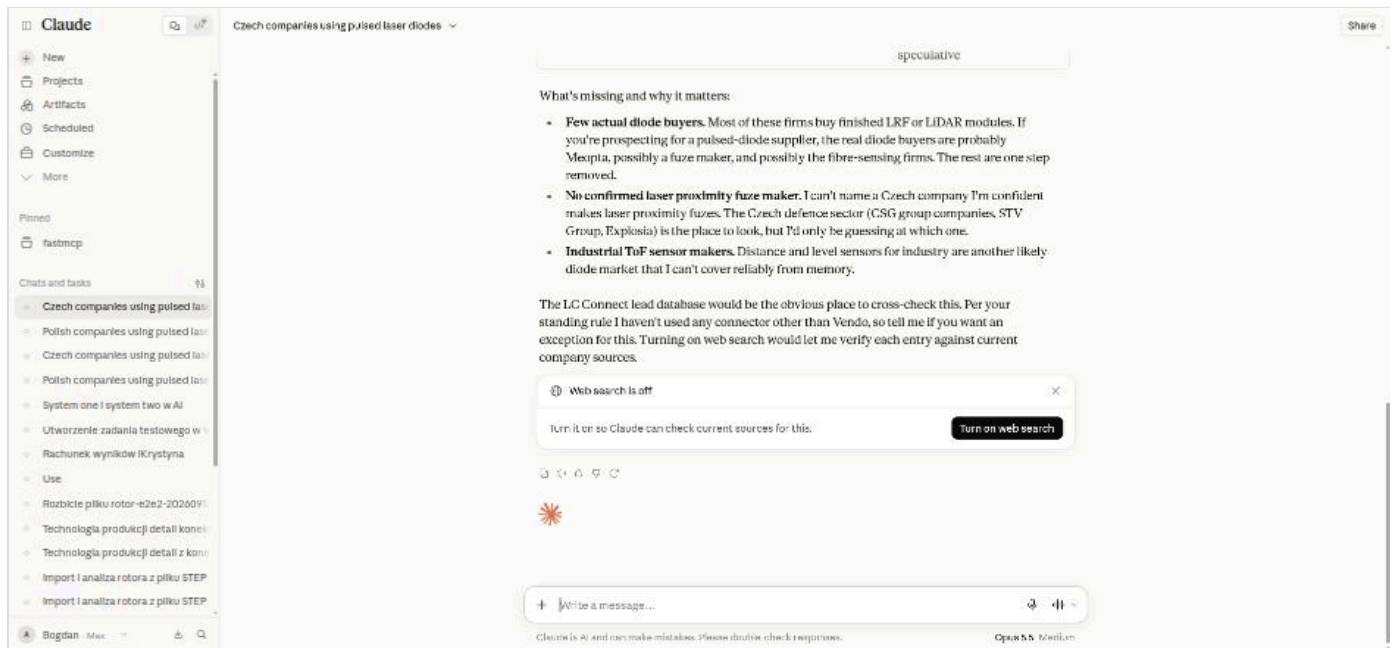
Opus 55 Medium

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Przetwarzano przez 31s

I found a relatively small number of **Polish companies for which there is public evidence connecting an actual product/application with 905 nm or 1550 nm pulsed laser technology**. The public documentation is much better for 905 nm than for 1550 nm.

One important distinction: **"905 nm laser" does not by itself prove that the source is a pulsed laser diode (PLD)**. For several products the wavelength is public but the exact emitter is not. I therefore give a separate confidence level for the PLD identification.

Company	City	Product / application	Wavelength	Evidence for pulsed diode
CRW Telesystem-Mesko	Lubiszów / Stare Babice	Laser rangefinder modules; optoelectronic systems; Piorun proximity-fuze work	905 nm	Strong – company describes a semiconductor laser-diode rangefinder module; ToF ranging implies pulsed operation although the public material doesn't explicitly call it 905 nm
TOYA S.A. / YATO	Wrocław	YATO YT-73129 / YT-731290 laser rangefinders	905 nm	Wavelength and ranging are documented, but the exact diode type/pulse characteristics aren't published
AiQ robotics P.S.A.	Pniewy	C16/C32 3D LIDAR scanners for mobile robotics	905 nm	Strong for pulsed operation – manufacturer/distributor "impulse" detection at 905 nm; exact PLD component identified
MESKO S.A.	Skarżysko-Kamienna	PIORUN MANPADS, incorporating laser proximity fuze	Laser wavelength not publicly	The proximity fuze is explicitly laser/optoelectronic; documentation verify 905/1550 nm for the production fuze

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Chat

+ Work

Od czego zaczniemy?

+

Zapytaj ChatGPT

Myślenie

Dodaj zdjęcia i pliki

Prześlij z komputera

Dodaj pliki z biblioteki

Przeglądaj i wyszukuj swoje pliki

Utwórz obraz

Wizualizuj, co chcesz

Pracuj w projekcie

Rozpocznij czat w projekcie

Rysuj

Narysuj i dołącz obraz

Wyszukiwanie w sieci

Wyszukuj wiadomości i informacje w czasie rzeczywistym

Myślenie

Aby uzyskać szczegółowe odpowiedzi

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Czech PLD Companies

Lista firm stosujących PLD

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Przetwarzano przez 27s

I found a **small number of Czech companies with public evidence of 905/1550 nm laser-ranging products**. The strongest matches are in military/security electro-optics rather than automotive LiDAR.

Company	City	Product / application	Wavelength	Confide
ARCHER-LPP s.r.o.	Pardubice / Prague	Thermal/night-vision sights and multisensor binoculars with integrated LRF; e.g. Archer TSA-7 (5 km) and MSB-8	1550 nm; some products also 905 nm	Very high
SEFOR Solutions s.r.o.	Prague	SEFOR MSBi multispectral binocular for reconnaissance, surveillance, border patrol and targeting, with integrated LRF	905/1550 nm, up to 8 km	High
OptiDyner s.r.o. / Night Pearl	Bystřice pod Hostýnem	Night Pearl thermal sights with integrated LRF, e.g. FOX 35 ELITE II LRF	905 nm, up to 1 km	Medium
Abacus Electric / EVOLVEO	Planá / České Budějovice	EVOLVEO Range Finder 1K , hunting/target rangefinder	905 nm, 5–1000 m	Medium-low
FOMEI s.r.o.	Hradec Králové / Prague	Danubia DJE 600 rangefinder sold under the FOMEI polio	905 nm	Low-medium

Powrót do aplikacji

Wyszukaj

Osobiste

Ogólne

Wygląd

Powiadomienia

Profil

Zabezpieczenia i logowanie

Konto

Kontrola rodzicielska

Zaufana osoba

Głos

Pamięć

Personalizacja

Zwierzęta

Klawiatura

Użycie

Płatności

Wtyczki

Przeglądaj katalog

Zarządzaj wtyczkami, połączonymi kontami i uprawnieniami

Wyszukaj wśród zainstalowanych wtyczek

Domyślne uprawnienie

Zezwalaj na narzędzia niskiego ryzyka

Dowiedz się więcej o uprawnieniach wtyczek

Sites

Build interactive site previews

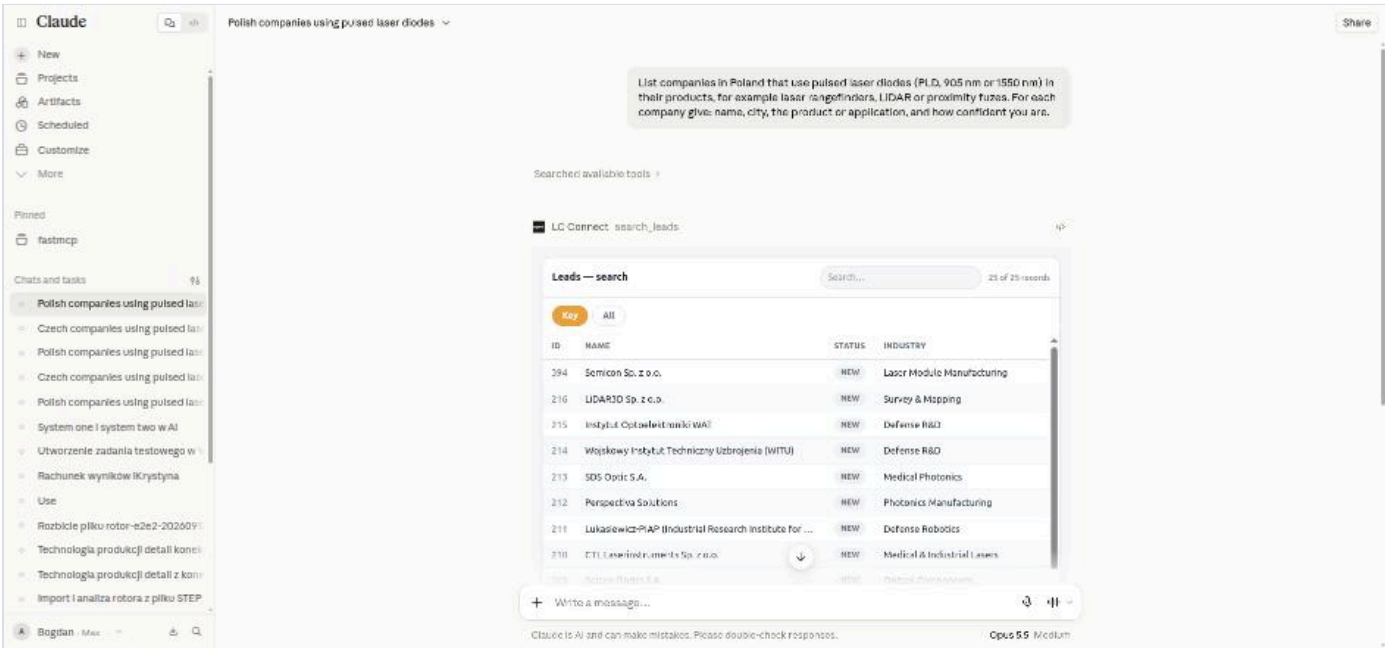
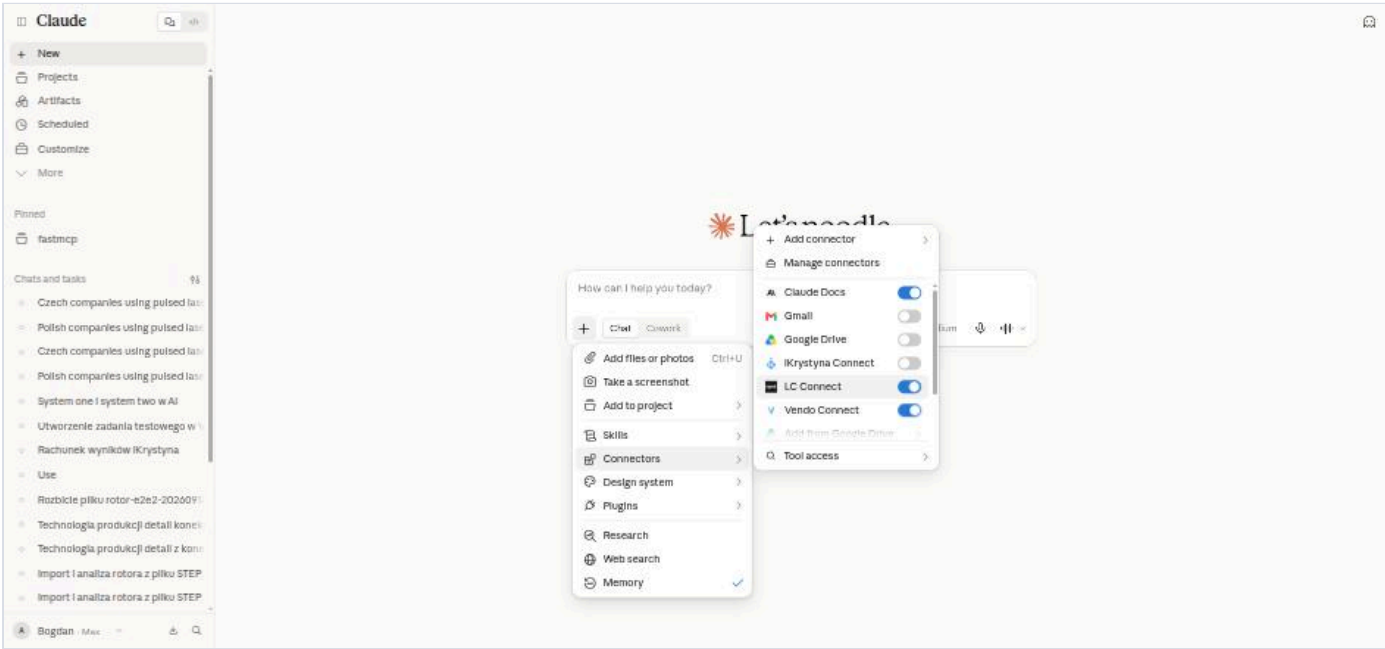
Vendo Connect

Vendo Connect

Zezwalaj na narzędzia niskiego ryzyka

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v2 runs used in this document (connector actually invoked)



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Czech companies using pulsed laser

Polish companies using pulsed laser

Czech companies using pulsed laser

Polish companies using pulsed laser

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Rachunek wyników IKrystyna

Use

Rozbicie pliku rotor-e2e2-202609

Technologia produkcji detali kone

Technologia produkcji detali z kon

Import i analiza rotora z pliku STEP

Bogdan

Mac

Polish companies using pulsed laser diodes

214	Współowy Instytut Techniczny uzbrojenia (WITU)	NEW	Defense R&D
213	SDS Optic S.A.	NEW	Medical Photonics
212	Perspectiva Solutions	NEW	Photonics Manufacturing
211	Lukasiewicz-PIAP (Industrial Research Institute for ...	NEW	Defense Robotics
210	CTL Laser Instruments Sp. z o.o.	NEW	Medical & Industrial Lasers
209	Solaris Optics S.A.	NEW	Optical Components
208	Opt Lasers (Tomorrow's System Sp. z o.o.)	NEW	Industrial Laser Systems
207	DIAM Tech Sp. z o.o.	NEW	Laser Electronics
206	Huta Stalowa Wola S.A. (HSW)	NEW	Defense Vehicles & Turrets
205	Zakłady Mechaniczne Tarnów S.A. (ZMT)	NEW	Defense Weapons Systems
204	Autocomp Management Sp. z o.o.	NEW	Defense Simulators

All 25 records

Download CSV

I.C. Connect returned 25 Polish leads (tool: `search_leads`, filter country = Poland), shown in the table above; it filters by country only, not by P.I.D use, so check the product/application column, and tell me if city or confidence are missing and I'll pull them per lead.

+

Write a message...

🗣️

🔊

🔍

Claude is AI and can make mistakes. Please double-check responses.

Opus 55 Medium

Claude

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Czech companies using pulsed laser

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Bogdan

Mac

How can I help you today?

+

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Codeview

Add files or photos

Take a screenshot

Add to project

Skills

Connectors

Design system

Plugins

Research

Web search

Memory

+

Add connector

Manage connectors

Claude Docs

Gmail

Google Drive

IKrystyna Connect

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Polish companies using pulsed laser

Czech companies using pulsed laser

Polish companies using pulsed laser

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Utworzenie zadania testowego w

Rachunek wyników IKrystyna

Use

Rozbicie pliku rotor-e2e2-202009

Technologia produkcji detail konek

Technologia produkcji detail z kon

Bogdan

Max

Czech companies using pulsed laser diodes

Share

Leads — search

Search...

20 of 29 records

KeyAll

ID	NAME	STATUS	INDUSTRY	COUNTRY
237	Ray Service a.s.	NEW	Defense Electronics / Cable Harnesses	CZE
236	Lightigo s.r.o.	NEW	Laser Spectroscopy / Measurement	CZE
235	LASER TECH spol. s r.o.	NEW	Industrial Laser Processing	CZE
234	Saab Czech Republic (Training Systems)	NEW	Defense / Training Simulation	CZE
233	URC Systems spol. s r.o.	NEW	Defense / Electronic Warfare	CZE
232	IPG Photonics (Czech Republic) s.r.o.	NEW	Fiber Lasers / Industrial Laser Systems	CZE
231	Opacilio s.r.o.	NEW	Holographic Security / Optics	CZE
230	Aero Vokochody Aerospace a.s.	NEW	Defense / Aerospace	CZE
229	Tatra Defence Vehicle a.s.	NEW	Defense / Armored Vehicles	CZE
228	Excalibur Army spol. s r.o.	NEW	Defense / Armored Vehicles	CZE
227	VULIS a.s. Liberec	NEW	Research & Engineering / Laser Applications	CZE
226	Narran s.r.o.	NEW	Industrial Laser Systems	CZE
225	GenEvo s.r.o.	NEW	Automotive Electronics / Laser Detection	CZE
223	Micro-Epsilon Czech Republic s.r.o.	NEW	Industrial Sensors / Laser Measurement	CZE

All 29 records

Download CSV

Write a message...

Opus 55

Medium

Claude is AI and can make mistakes. Please double-check responses.

Claude

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fastmcp

Chats and tasks

Czech companies using pulsed laser

Polish companies using pulsed laser

Czech companies using pulsed laser

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Technologia produkcji detail konek

Technologia produkcji detail z kon

Bogdan

Max

Czech companies using pulsed laser diodes

Share

233

URC Systems spol. s r.o. NEW | Defense / Electronic Warfare | CZE || 232 | IPG Photonics (Czech Republic) s.r.o. | NEW | Fiber Lasers / Industrial Laser Systems | CZE |
231	Opacilio s.r.o.	NEW	Holographic Security / Optics	CZE
230	Aero Vokochody Aerospace a.s.	NEW	Defense / Aerospace	CZE
229	Tatra Defence Vehicle a.s.	NEW	Defense / Armored Vehicles	CZE
228	Excalibur Army spol. s r.o.	NEW	Defense / Armored Vehicles	CZE
227	VULIS a.s. Liberec	NEW	Research & Engineering / Laser Applications	CZE
226	Narran s.r.o.	NEW	Industrial Laser Systems	CZE
225	GenEvo s.r.o.	NEW	Automotive Electronics / Laser Detection	CZE
223	Micro-Epsilon Czech Republic s.r.o.	NEW	Industrial Sensors / Laser Measurement	CZE

All 29 records

Download CSV

Write a message...

Opus 55

Medium

Claude is AI and can make mistakes. Please double-check responses.

The search_leads card above shows all 29 I.C. Connect leads with country "Czech", with their product, application and confidence fields. I didn't narrow it to pulsed-laser-diode users (905/1530 nm rangefinders, LiDAR, fuzes), so filter the table by product or application, or ask me to re-run the search with those terms.

✖

🔍

🔗

🔄

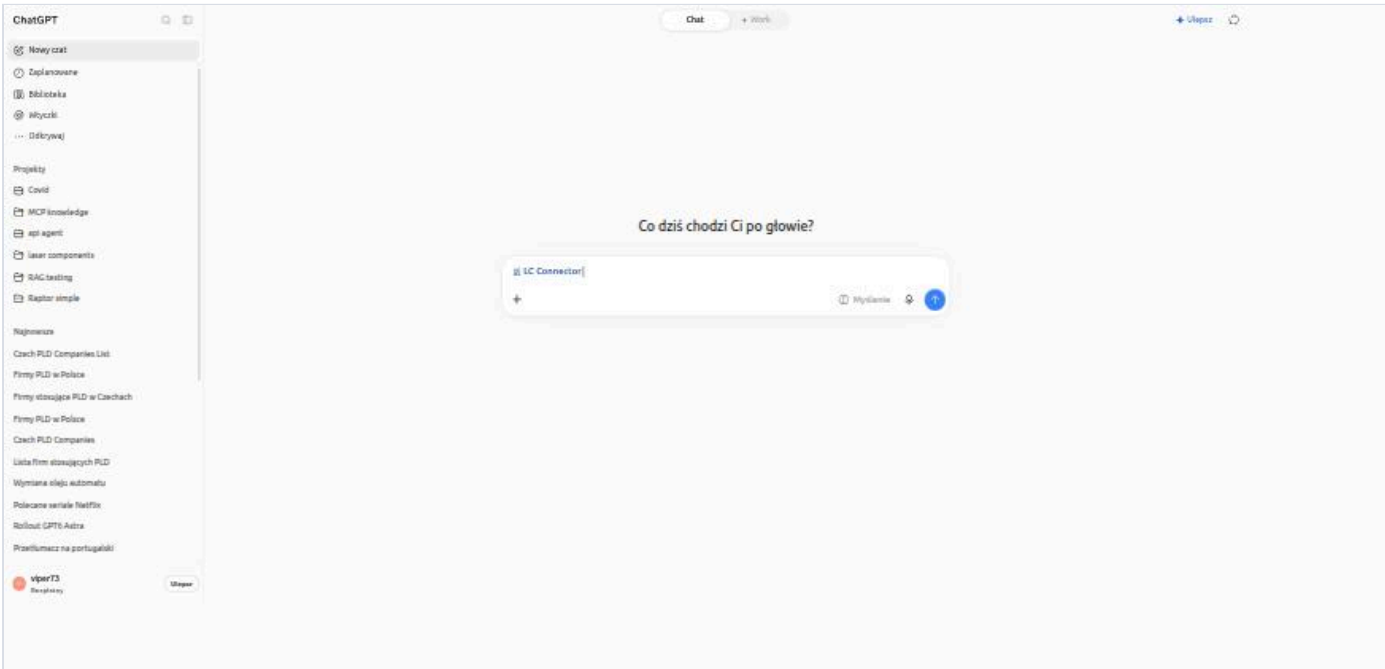
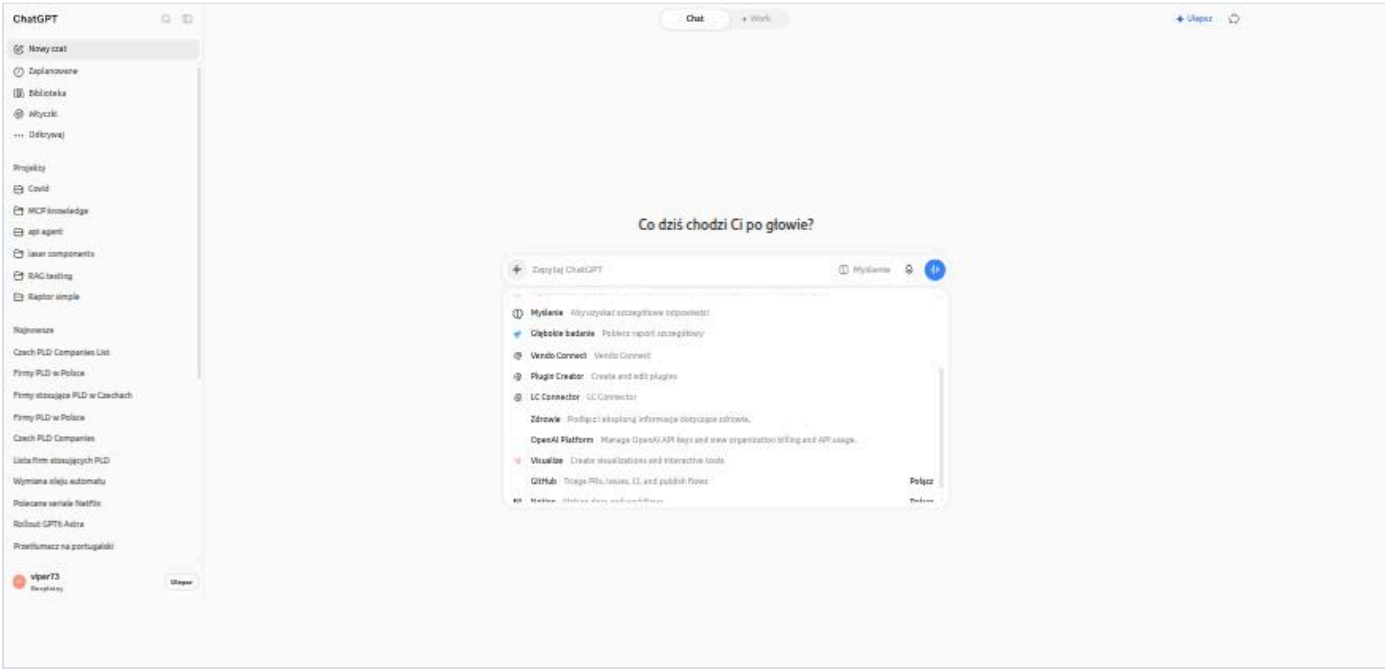
🗨️

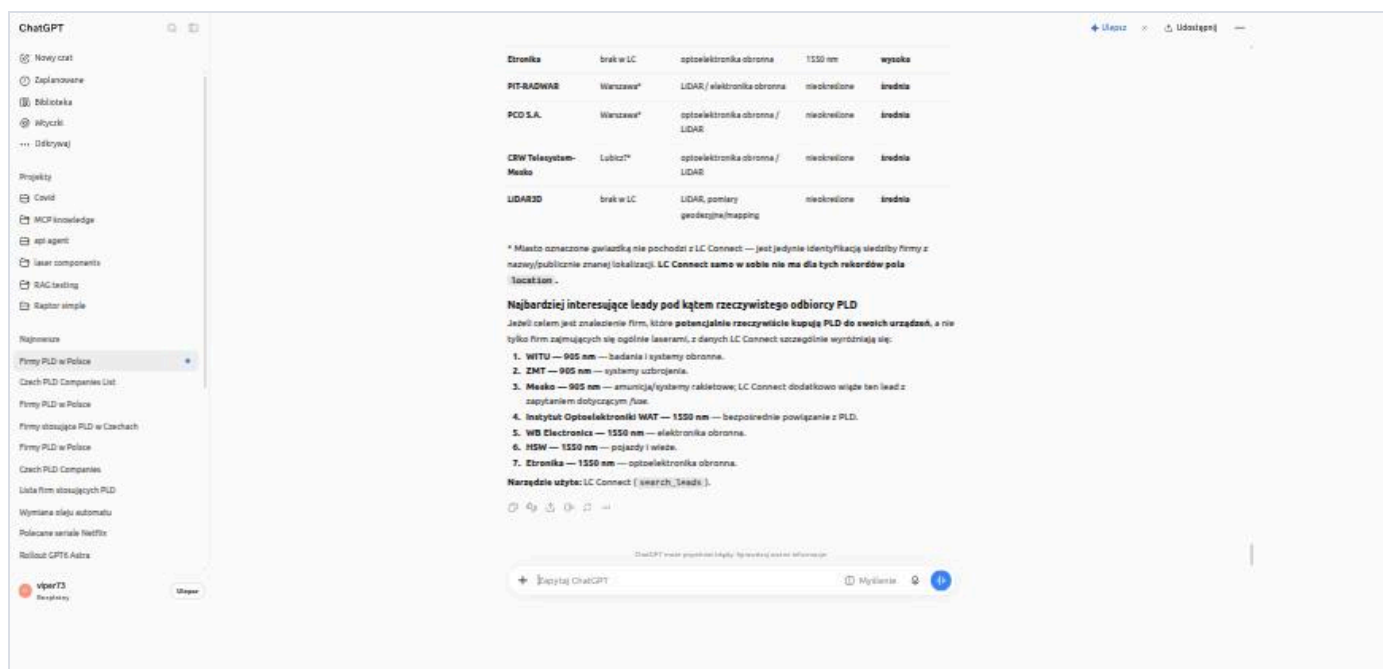
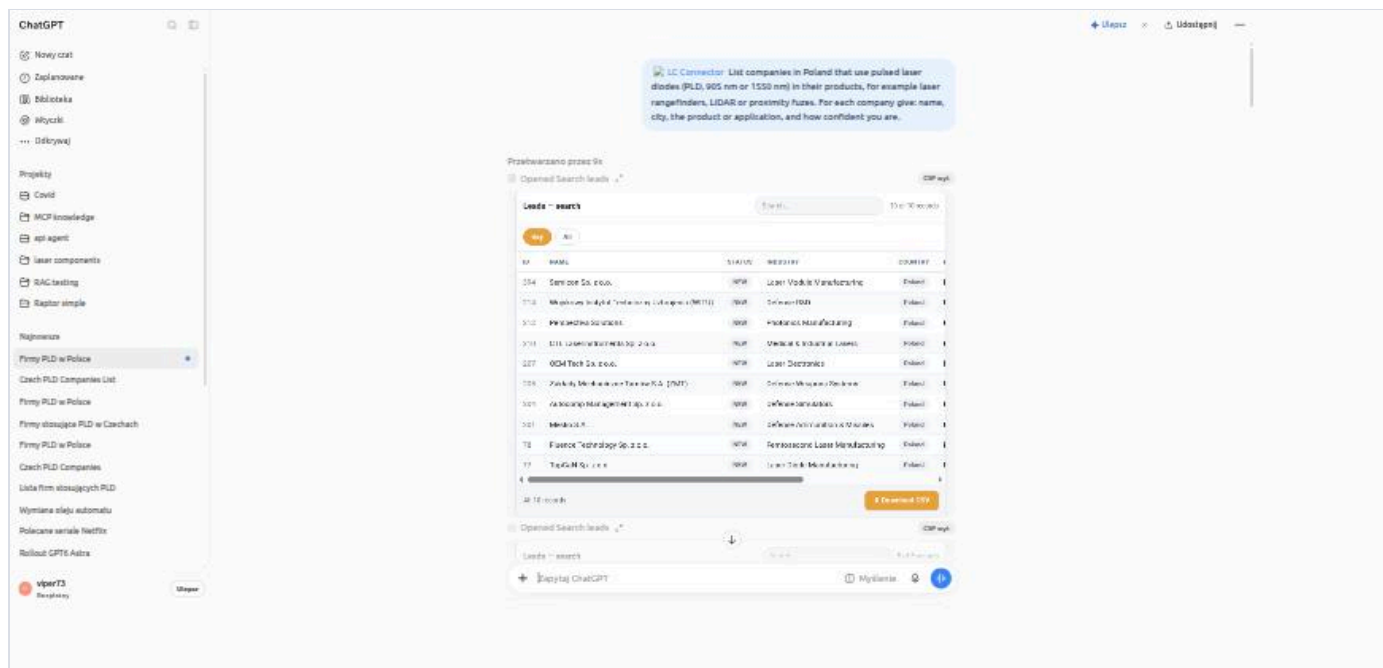
Just now

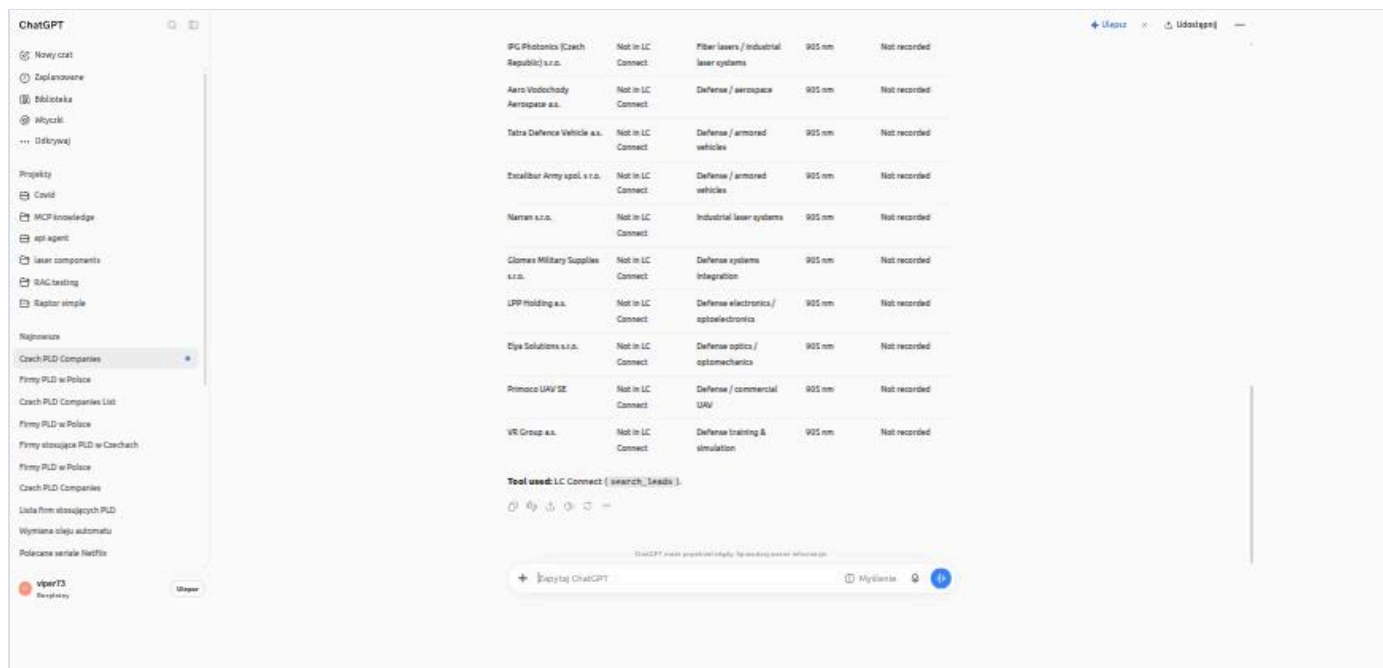
✖

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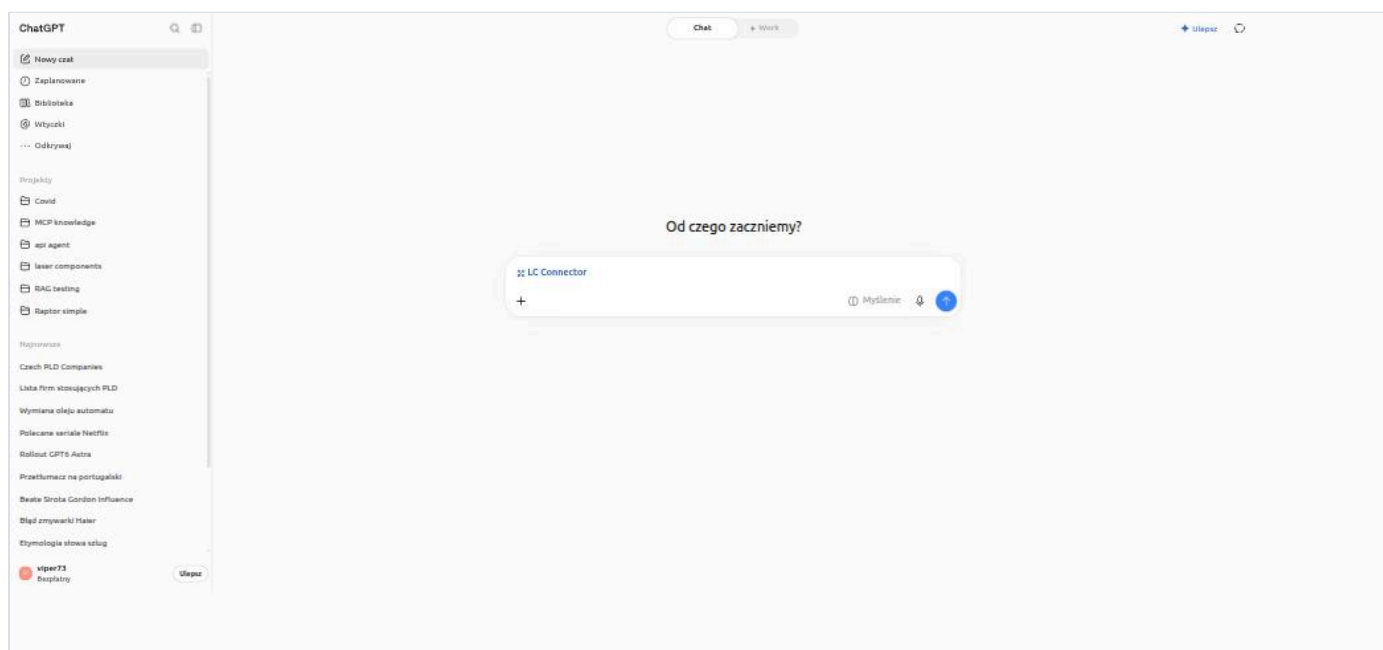
ChatGPT + LC reruns used in this document (22:43–22:46 CEST)

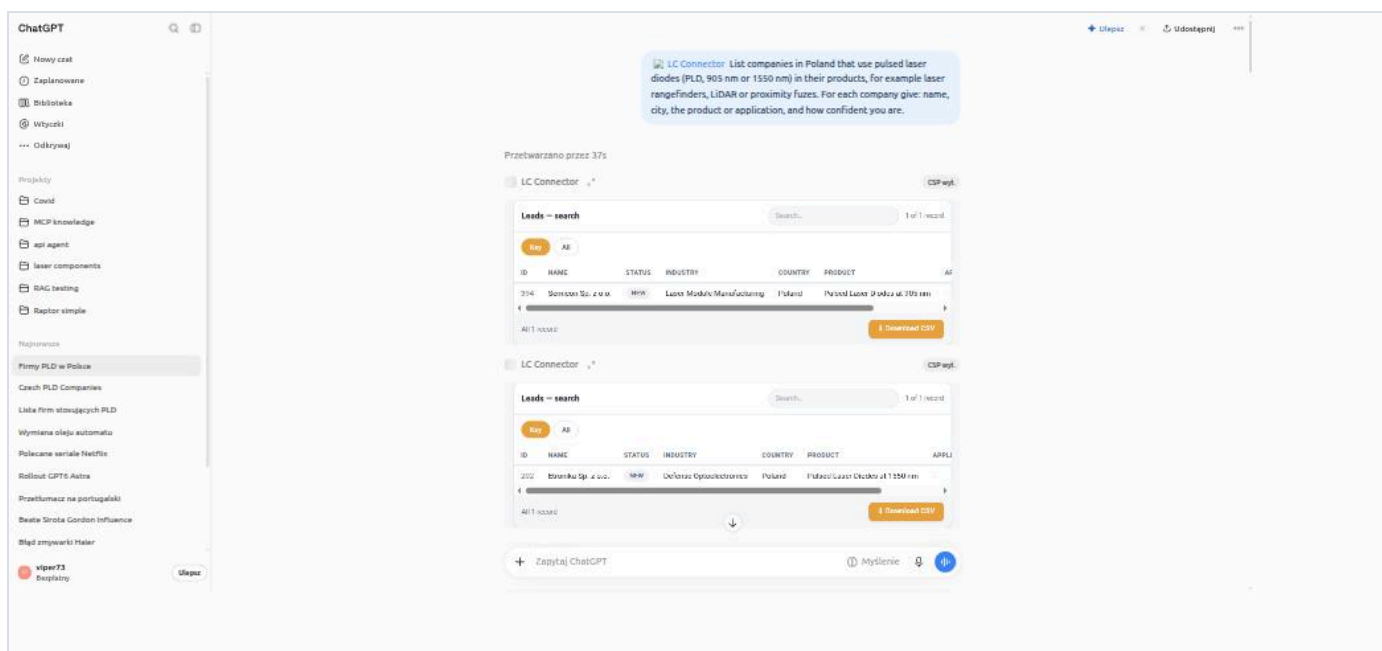
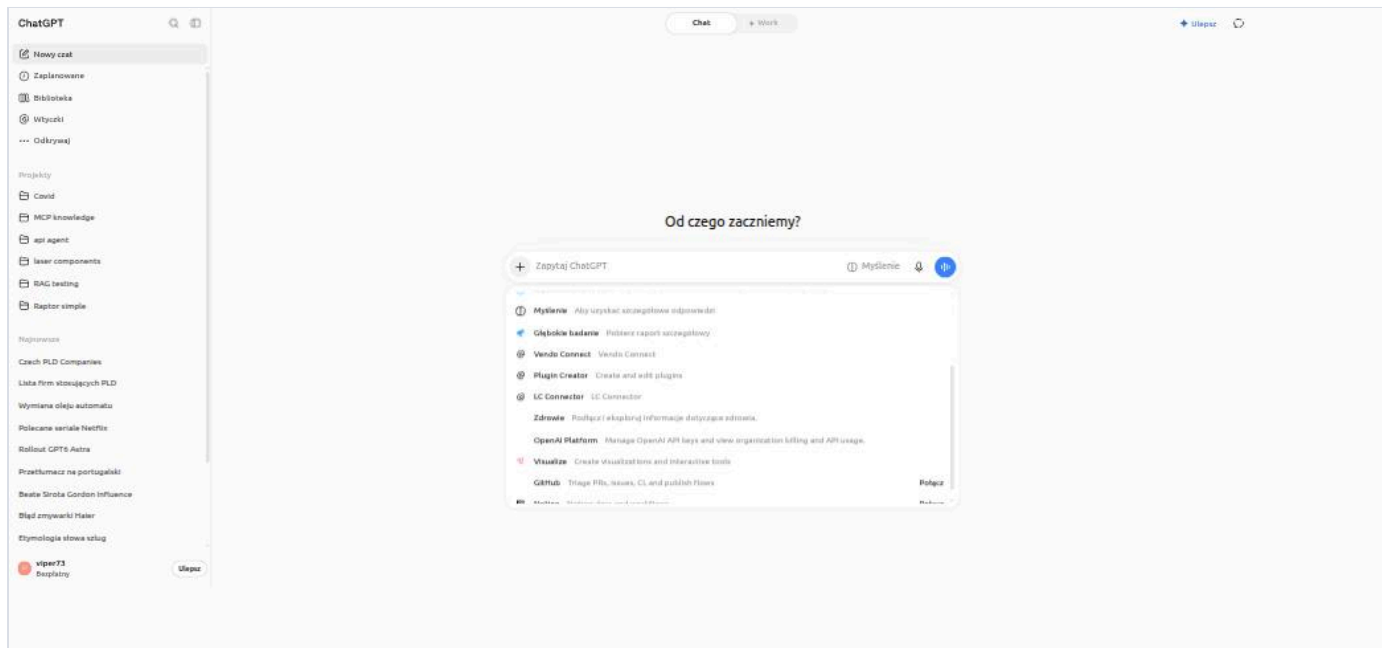


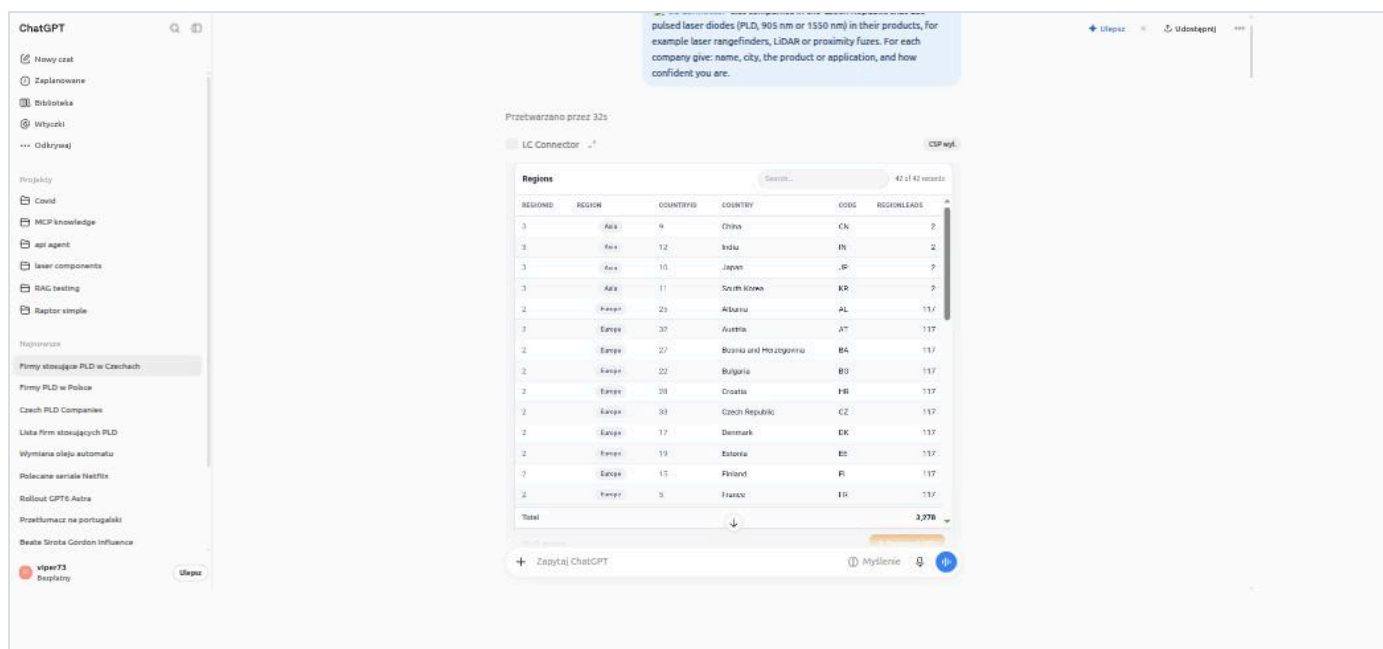
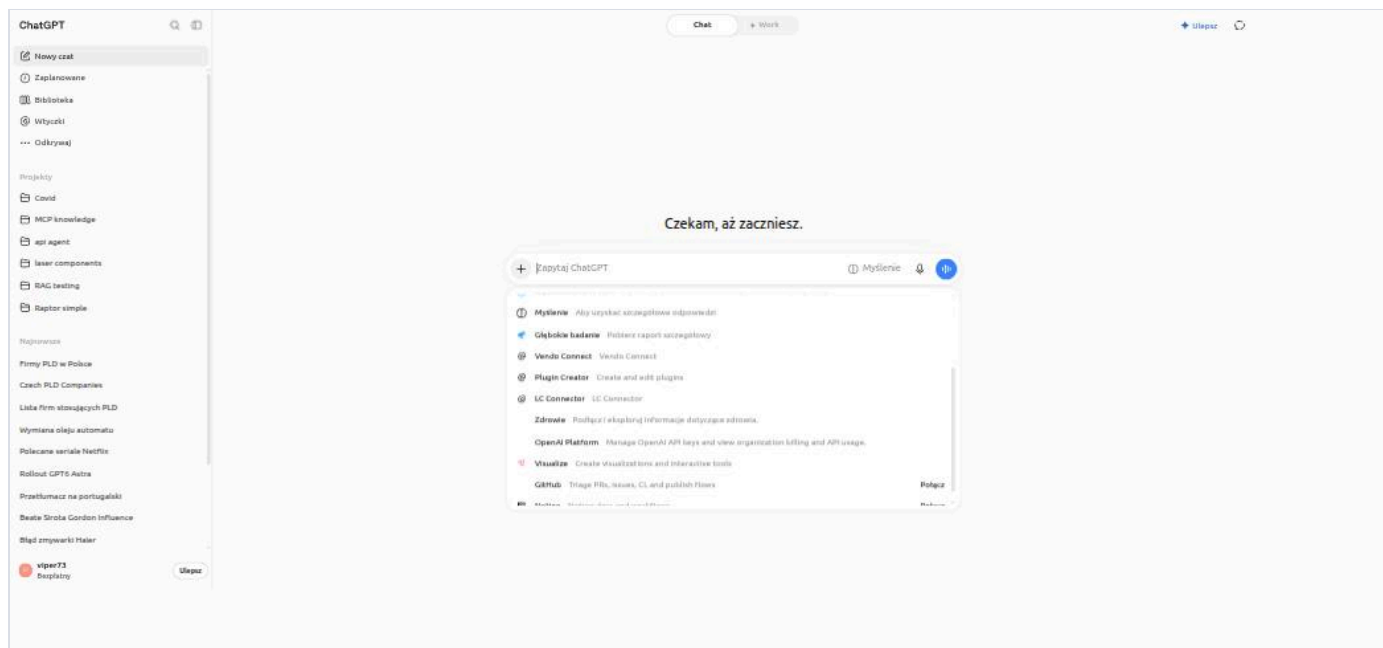


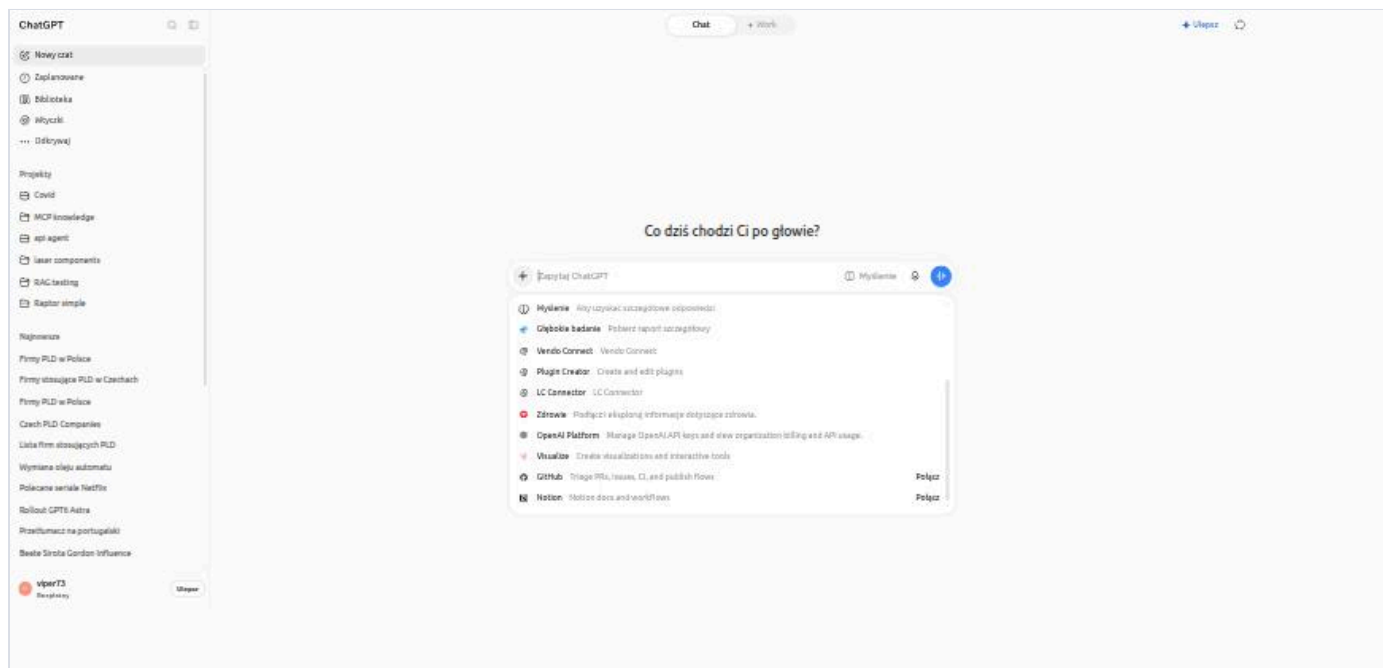


Superseded ChatGPT runs (Appendix A)

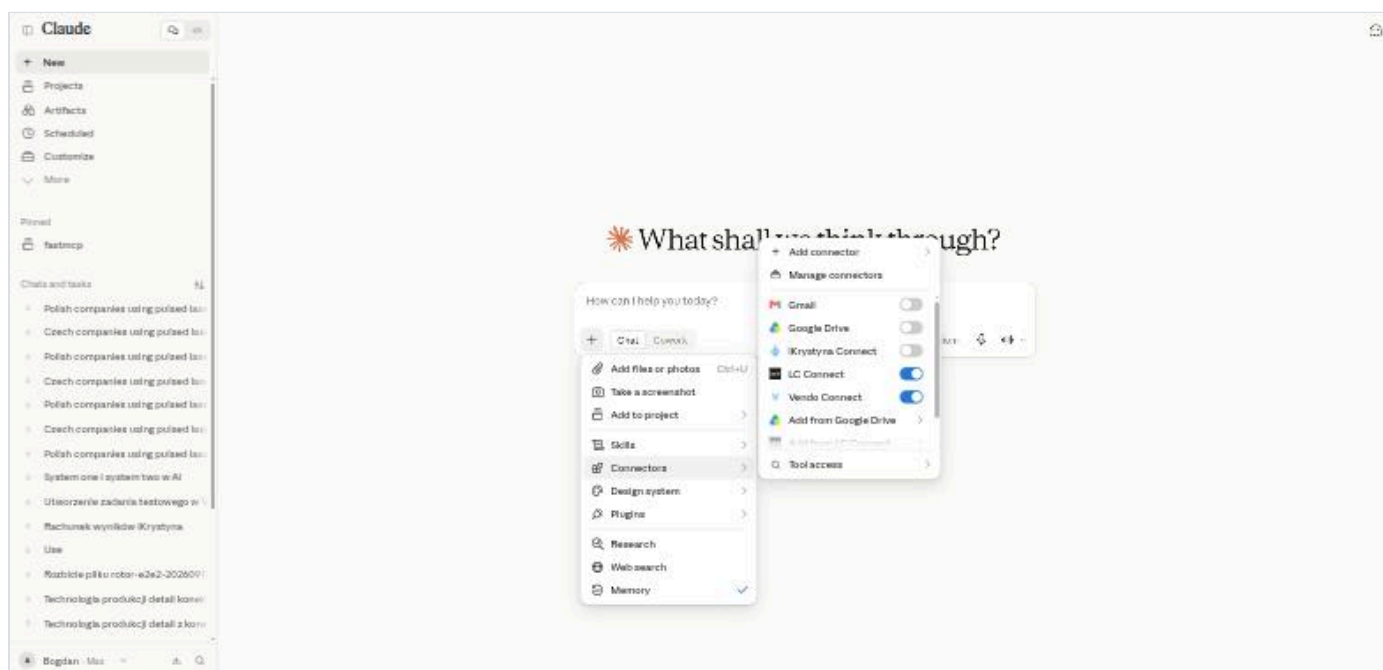








Post-fix Claude runs (Sections 3.4, 4.4 and 4.5)



Claude

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Chats and tasks

Polish companies using pulsed laser diodes

Polish companies using pulsed laser

Czech companies using pulsed laser

Polish companies using pulsed laser

Czech companies using pulsed laser

Polish companies using pulsed laser

Czech companies using pulsed laser

Polish companies using pulsed laser

System one i system two w AI

Utworzenie zadania testowego w

Machunek wyników IKrytyna

Use

Możliwość pliku robot: e2e2-20250V1

Technologia produkcji detail konw

Polish companies using pulsed laser diodes

share

Perspectiva Solutions	805 nm	?	Photonics manufacturing	Low
Solaris Optics S.A.	1550 nm	Nadzwym	Optical components, not light sources	Low
FTBRAIN	digitalized P.D	Zawernie (Rzeszów)	Fibre optics, possibly OTDR type test gear	Low
Fluence Technology	905 nm	Warsaw	Femtosecond fibre lasers, so a different technology	Low
TopGaN	905 nm	Warsaw	GaN laser diode maker, more likely a competitor than a user	Low

Possible gap (general knowledge, unverified): PCO S.A. (Warsaw) is Poland's best-known maker of laser rangefinders and military observation optics, and it does not appear among those P.D leads. Leads without a country assigned are only partly searchable, though, so it may exist without a country tag.

Several rows (Fluence, TopGaN, Solaris) look like weak product matches. You may want to ask me to use `report_issue` about the missing city field, or `save_learning` to flag those matches.

Write a message...

Opus 1.5 (Vector)

Claude AI can make mistakes. Please double check responses.

Claude

New

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Artifacts

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Polish companies using pulsed laser

Czech companies using pulsed laser

Polish companies using pulsed laser

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Polish companies using pulsed laser

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Use

Możliwość pliku robot: e2e2-20250V1

Technologia produkcji detail konw

What shall we think through?

How can I help you today?

Chat / Connect

Add files or photos

Take a screenshot

Add to project

Skills

Connectors

Design system

Plugins

Research

Web search

Memory

Add connector

Manage connectors

Gmail

Google Drive

IKrytyna Connect

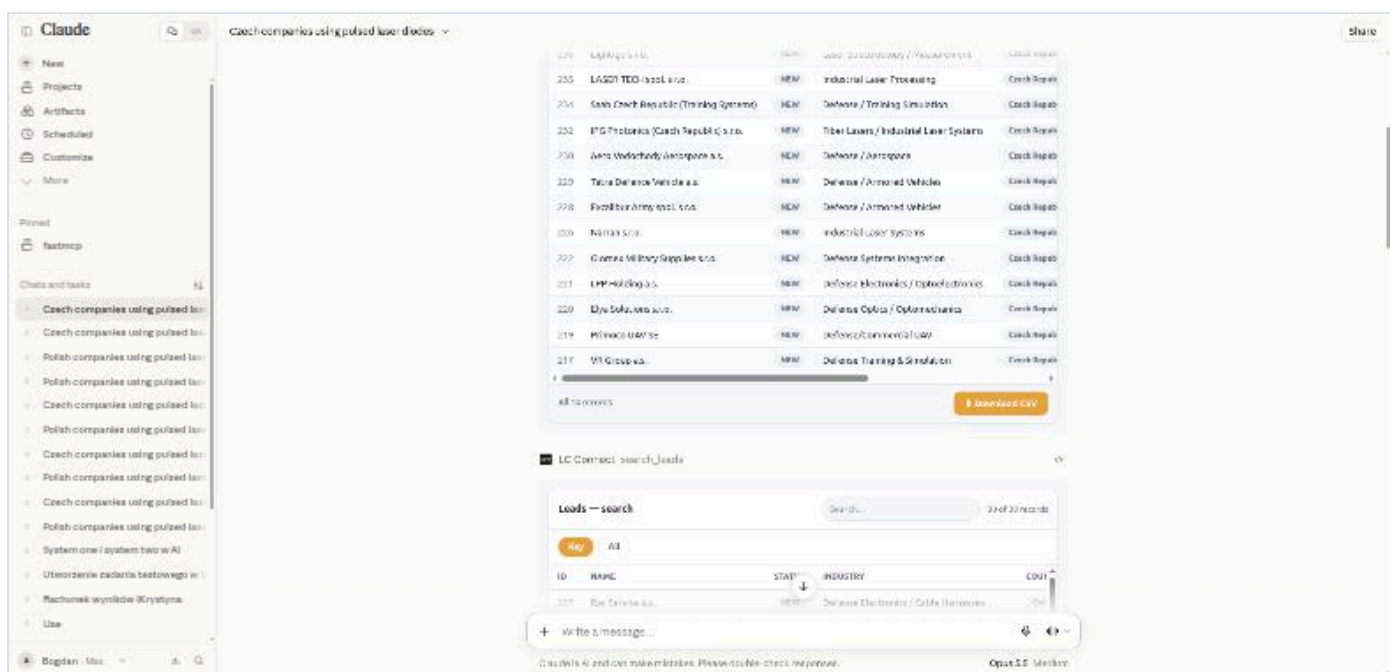
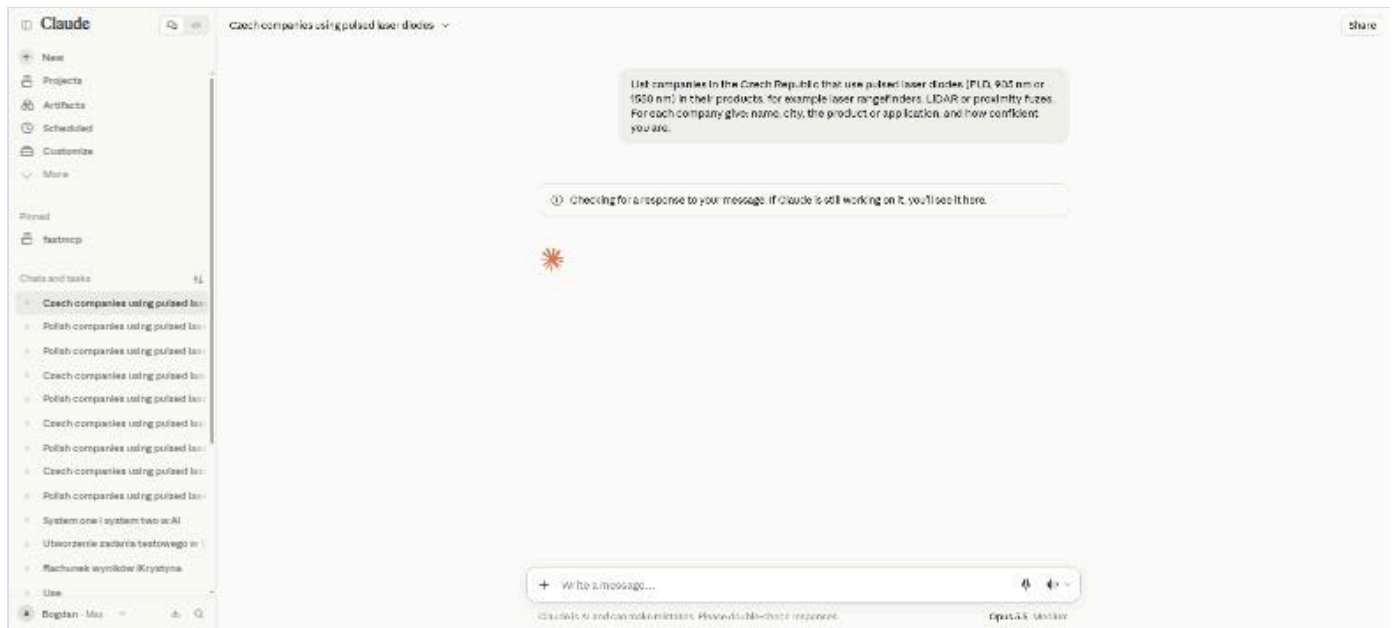
LC Connect

Vendo Connect

Add from Google Drive

Tool access

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Superseded session checks from the first draft, made with the automation profiles, not the owner's browser: [raw/session-check_*.png](#).